Chemistry of the Earth

Overview of the Semester

January Exam

Purpose of the exam:

- make sure that you thoroughly review the content of the Chemistry of the Earth module
- demonstrate your competency with the basic (geo)chemistry needed for future Earth and Environmental Science modules

The exam is based on content presented in lectures and practicals ONLY. There is no requirement that you have done any of the further reading that has been suggested throughout the semester.

Review ALL of the content throughout the semester: ALL lectures and practicals.

The exam is 2 hours in length. Extra time for those allowed it.

The exam is not open-book: you will not be able to refer to any notes or the internet.

You are allowed to use an approved calculator, but not a phone.

You will be given a periodic table.

Equations

I expect you to remember simple equations like:

- The equation for a partition coefficient

e.g. $^{Ni}D_{metal/silicate}$ = (concentration of Ni in metal) / (concentration of Ni in silicate)

I DO NOT expect you to memorise equations such as:

- Thermodynamic equations
- Equation for pH
- Stable isotope equations (α, δ)
- Radiogenic isotope equations (the isochron equation)

If there is a question relating to one of these I will give you the equation.

Example: An exam question might give you the equation:

$$\Delta G = \Delta H - T\Delta S$$

for a mineral reaction.

Then you might be given values of ΔG and ΔS, you will be asked to calculate the enthalpy change (ΔH) at a given temperature.

How to prepare for the exam

Review all of the lectures and practicals.

For the practicals, refer to the feedback documents to make sure you got answers right and that you understand the questions.

Read the Objectives at the beginning of each lecture and practical. Ask yourself if you have achieved these objectives.

Example: Practical 1, objective is:
"Understand which elements are the most abundant in the Solar System, and why"
Which elements are the most abundant? – you don't need to memorise a list, but if you know that the most abundant elements are a) low atomic number and b) have even numbers of protons, this is based on a better understanding of the process that forms atoms. Why are H and He very abundant in the Solar System? Why are C, N, O, Mg, Si abundant?

Chemistry of the Earth

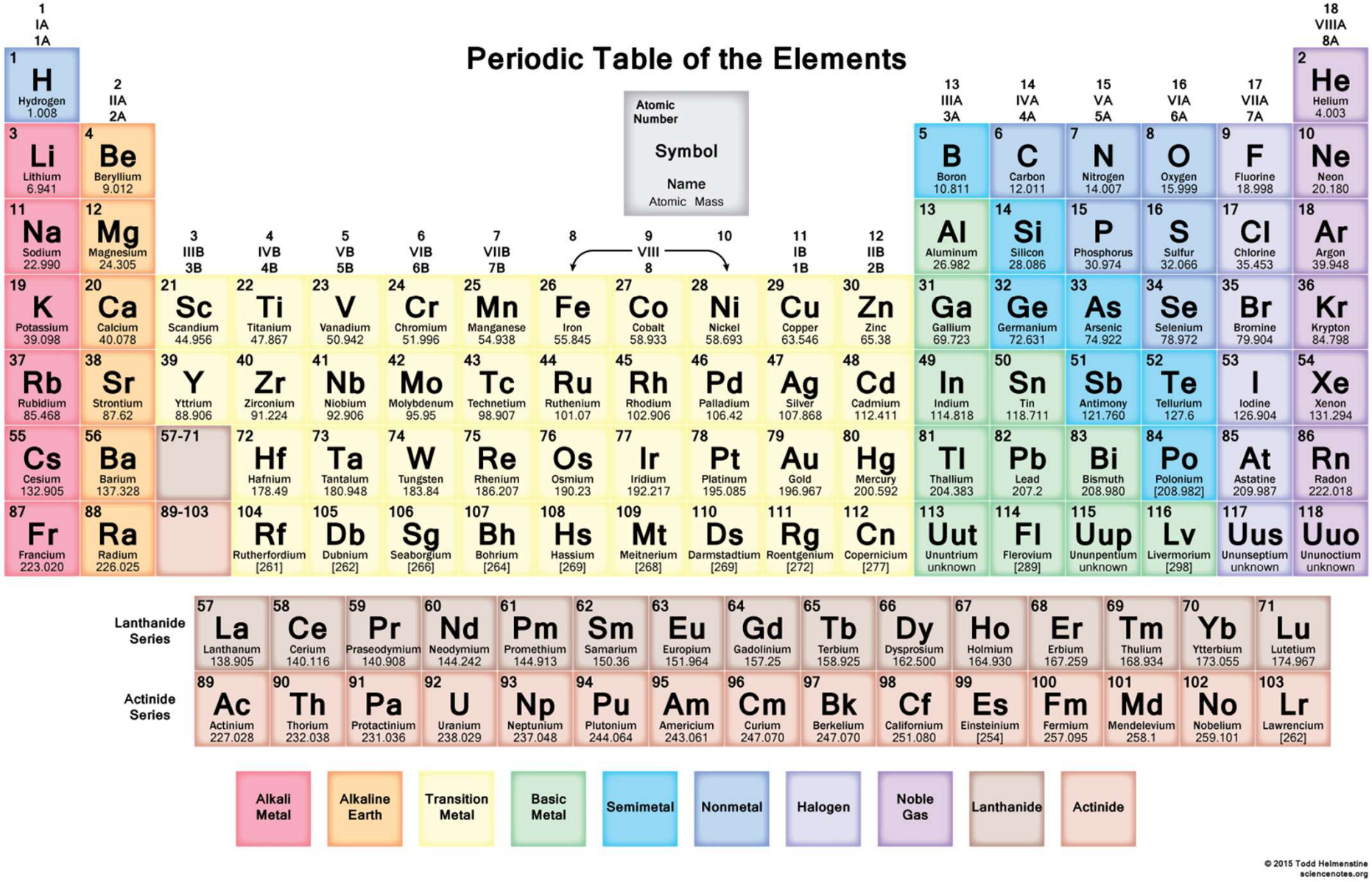
Lecture 1:
What is the Earth made of?

OBJECTIVES OF LECTURE

- Review the basic structure of atoms, elements, and isotopes
- Understand how chemical elements are produced in stars
- Consider how the chemistry of the Earth relates to its place in the Universe and the Solar System

OBJECTIVES OF PRACTICAL

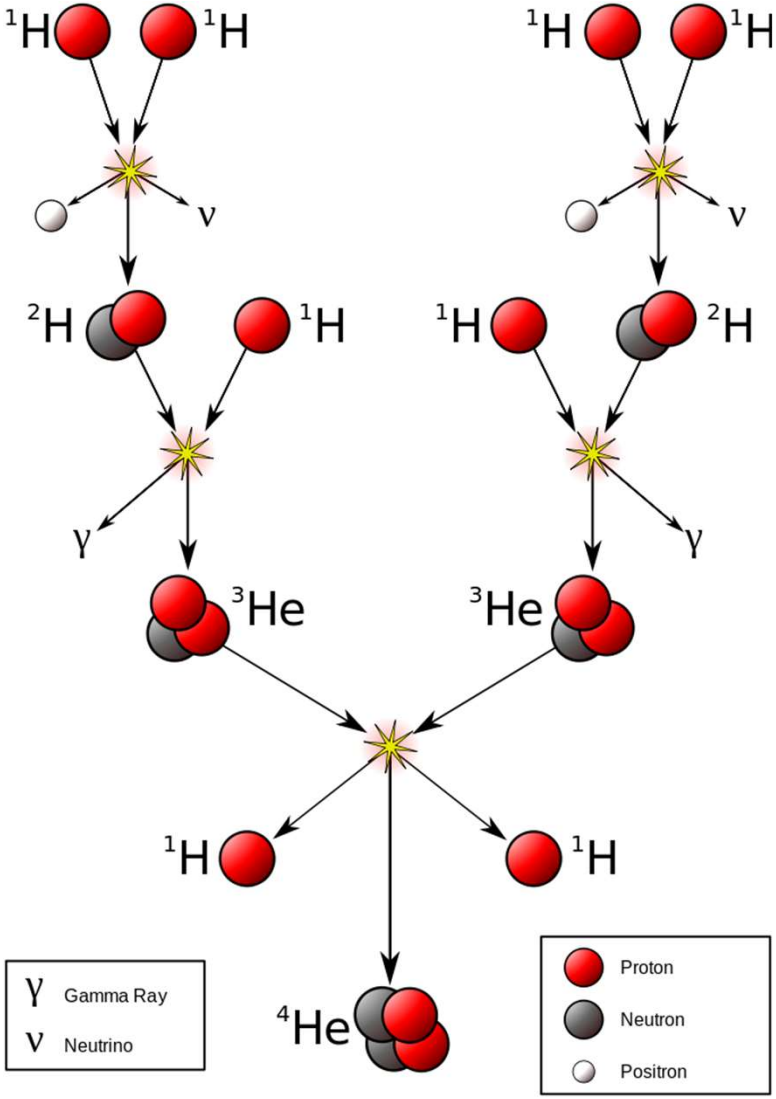
- Understand which elements are the most abundant in the Solar System, and why.
- Understand how the most abundant elements are distributed between the Earth's Core and Mantle.
- Calculate the chemistry of the whole Earth based on Core and Mantle compositions.
- Compare the composition of the whole Earth with the composition of the Solar System.



http://sciencenotes.org/printable-periodic-table/

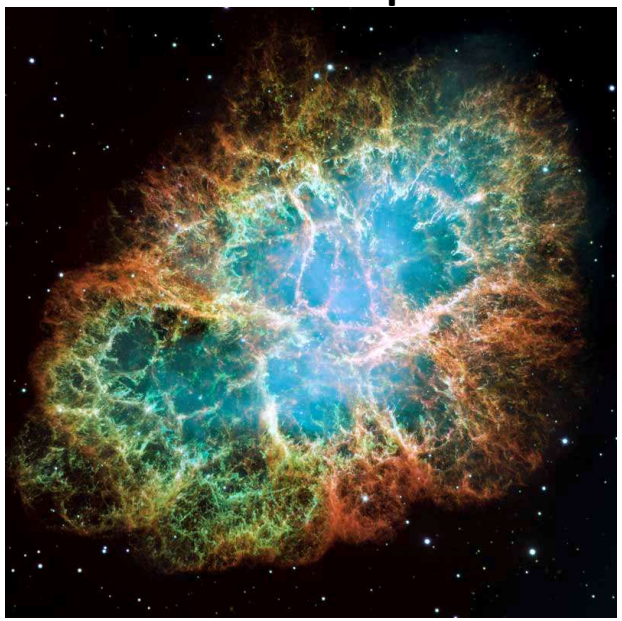
NUCLEOSYNTHESIS

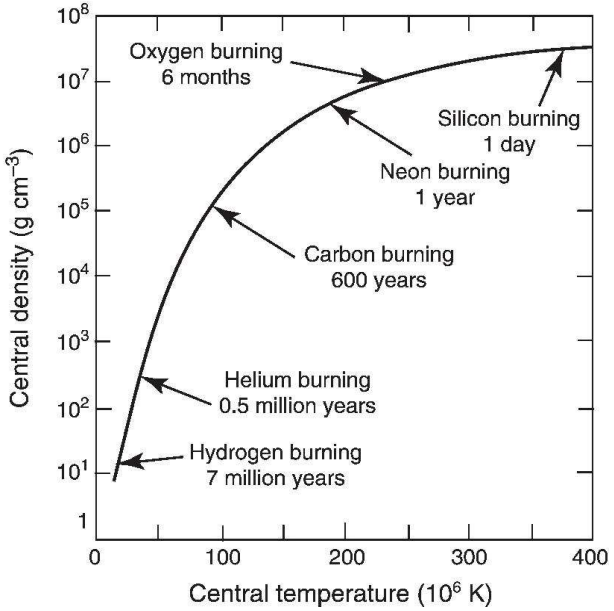
Hydrogen burning or proton-proton chain



Wikipedia

Supernova: neutron capture

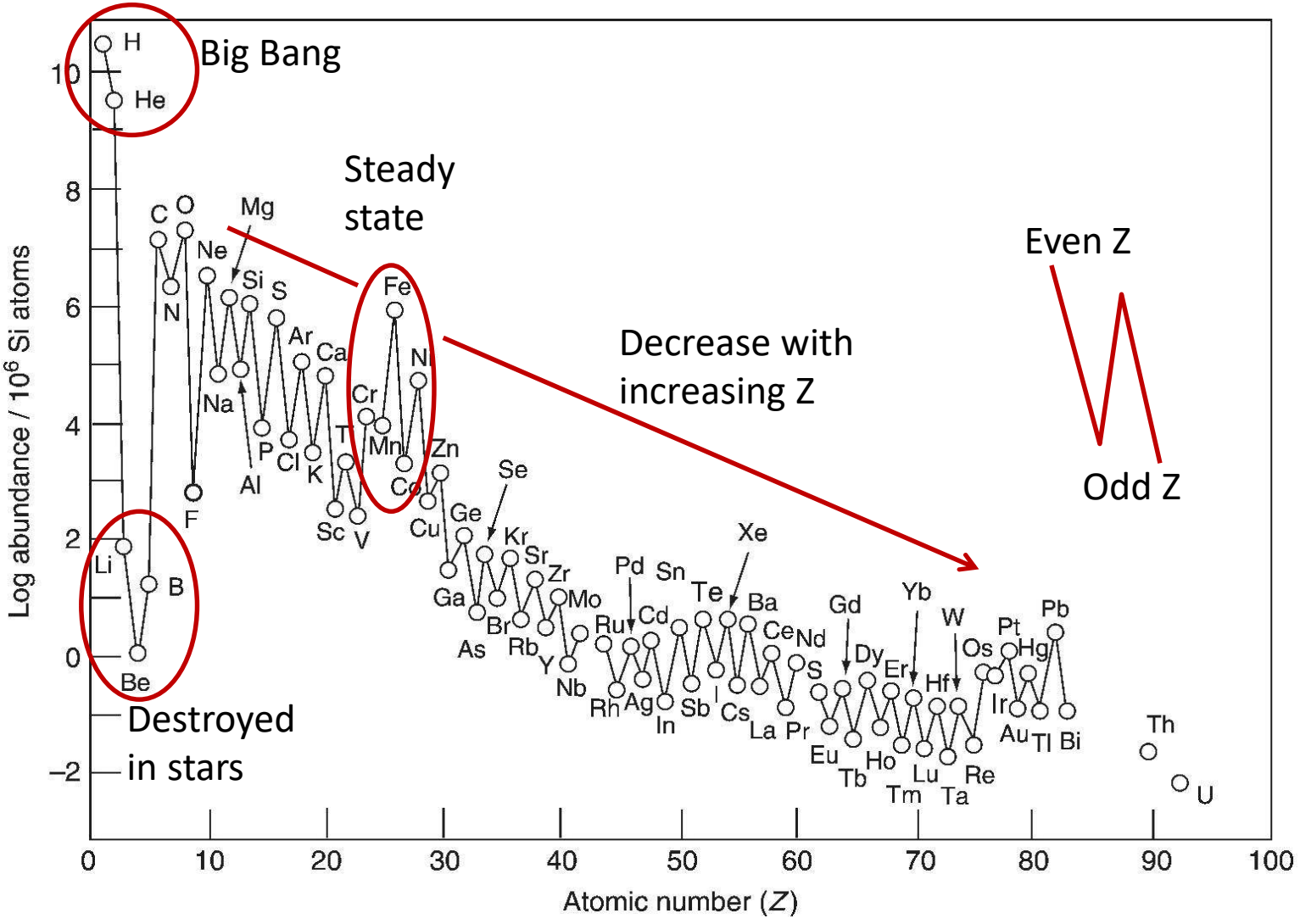




Introduction to Geochemistry: Principles and Applications, First Edition. Kula C. Misra.
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CHEMICAL COMPOSITION OF THE SUN AND SOLAR SYSTEM

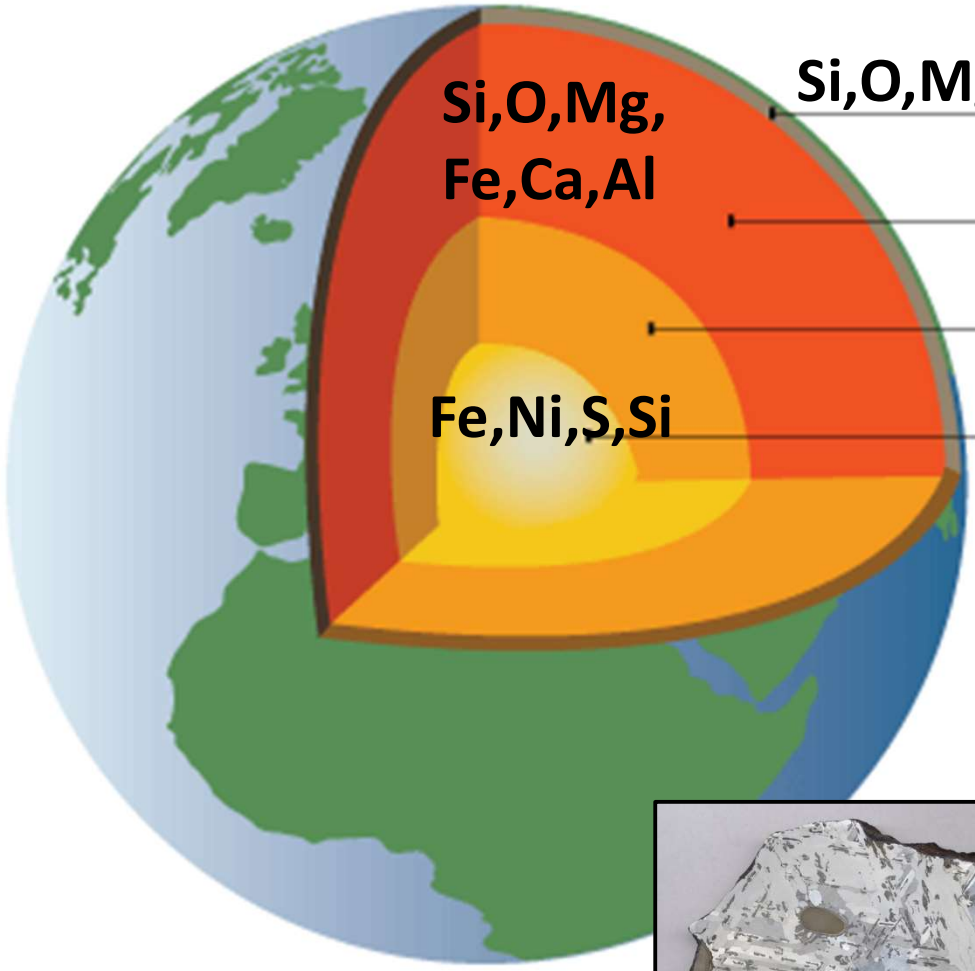
Composition makes sense when we know a little about nucleosynthesis.



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BULK COMPOSITION OF EARTH

Major elements in the different layers



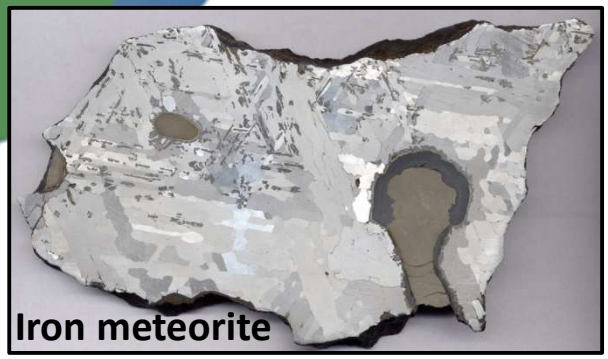
Si,O,Mg,Fe,Ca,Al,Na,K Crust

Mantle

Outer Core

Inner Core





Chemistry of the Earth

Lecture 2:

The Periodic Table – A Geochemist's View

OBJECTIVES OF LECTURE

- Review the Periodic Table of the Elements – Groups and Periods
- Understand electron configurations and why they are important
- Understand why groups of elements behave in similar ways
- Review the basic types of bonding important to minerals
- Introduction to the Geochemist's Periodic Table

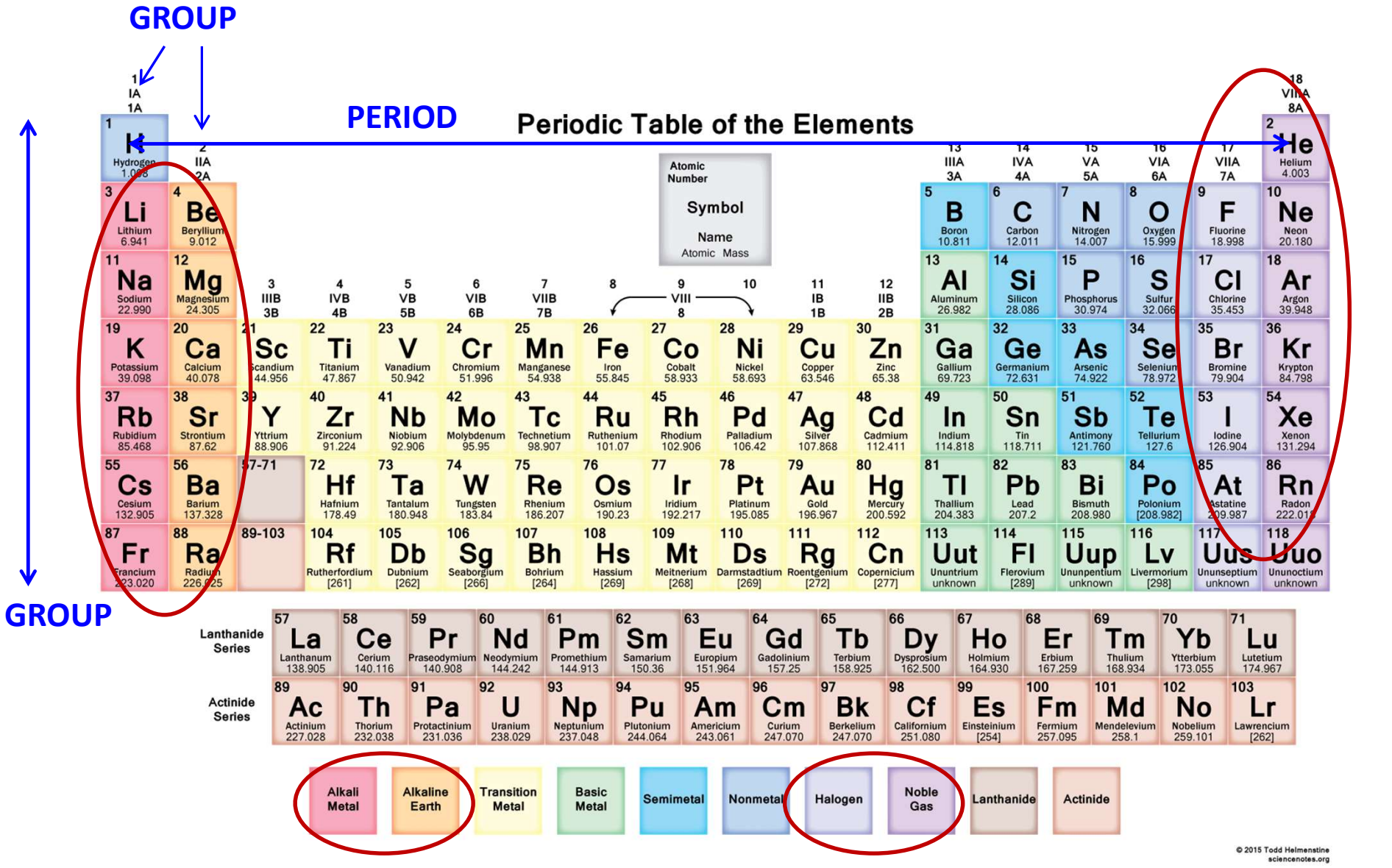
OBJECTIVES OF PRACTICAL

- Familiarize yourself with the Periodic Table, its groups and periods
- Review of atomic number, atomic mass, and isotopes
- Understand electron configurations and how they relate to the structure of the Periodic Table
- Practice writing mineral formulae based on charge balance

Periods are horizontal rows.

Groups are vertical columns - elements in groups are chemically similar.

Groups include: Alkali metal, Alkaline Earth, Halogens, Noble Gases



http://sciencenotes.org/printable-periodic-table/

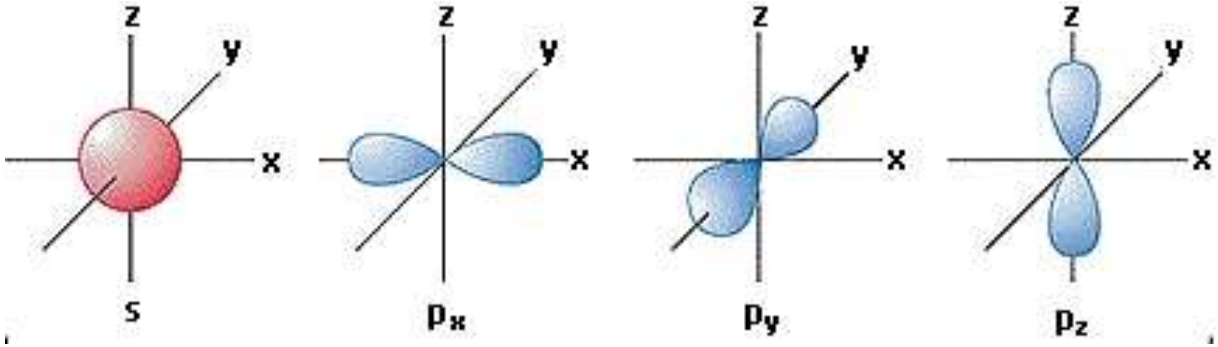
ELECTRON CONFIGURATION

The chemical properties of an element depend on the way its electrons are arranged around the nucleus – electron configuration

For a neutral atom, Number of electrons = Number of protons (= Z)

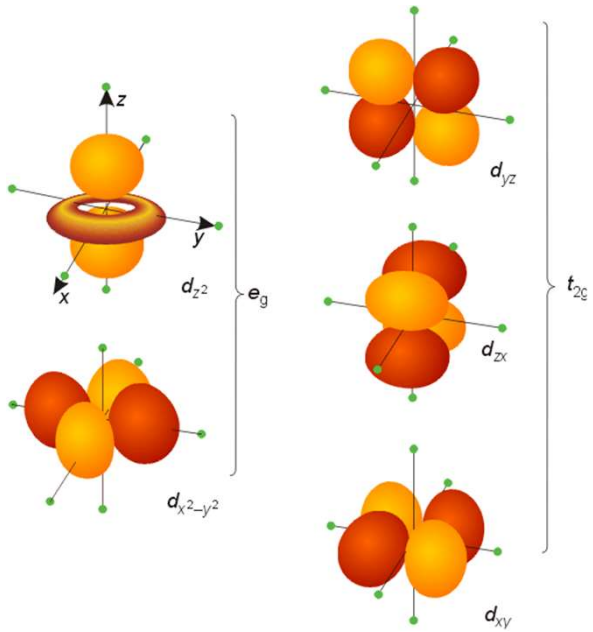
- Electrons are arranged in **SHELLS** around the nucleus
- Each shell has one or more **SUBSHELLS**
- Each subshell is composed of one or more **ORBITALS**
- Each orbital contains a maximum of two electrons

SHELL	SUBSHELL	Number of ORBITALS in subshell	Maximum number of ELECTRONS	Total electrons in shell
1 (K)	1s	1	2	2
2 (L)	2s	1	2	8
	2p	3	6	
3 (M)	3s	1	2	18
	3p	3	6	
	3d	5	10	
4 (N)	4s	1	2	32
	4p	3	6	
	4d	5	10	
	4f	7	14	



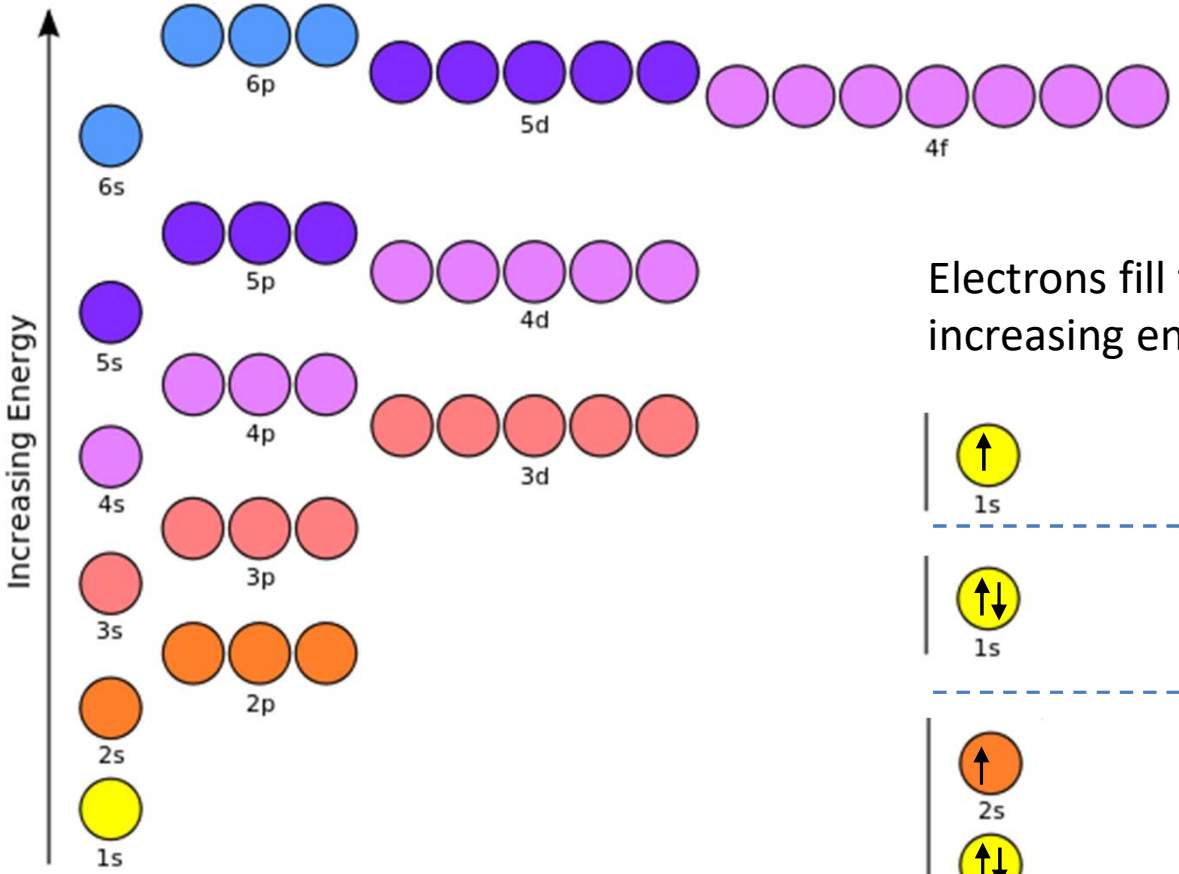
- **s orbitals** are spherical

- **p orbitals** consist of two lobes
- Each p orbital is aligned with a different axis (x, y, z)

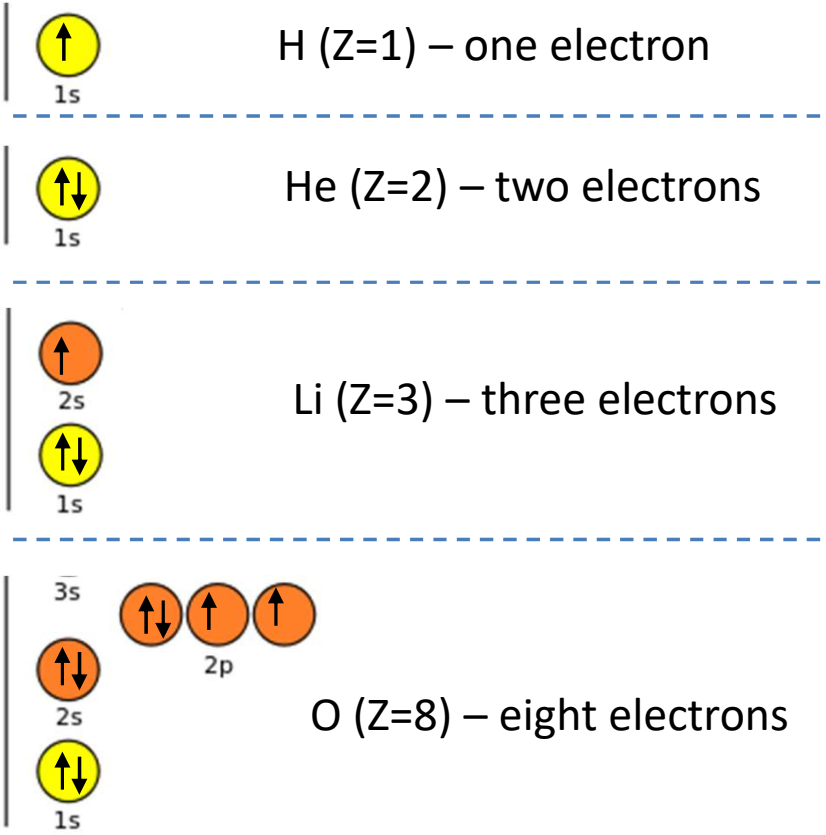


- Four sets of **d orbitals** consist of four lobes
- Each d orbital is aligned in a different direction relative to x, y, z axes
- The fifth d orbital (dz^2) has a unique shape

Electron orbitals have different energy levels:

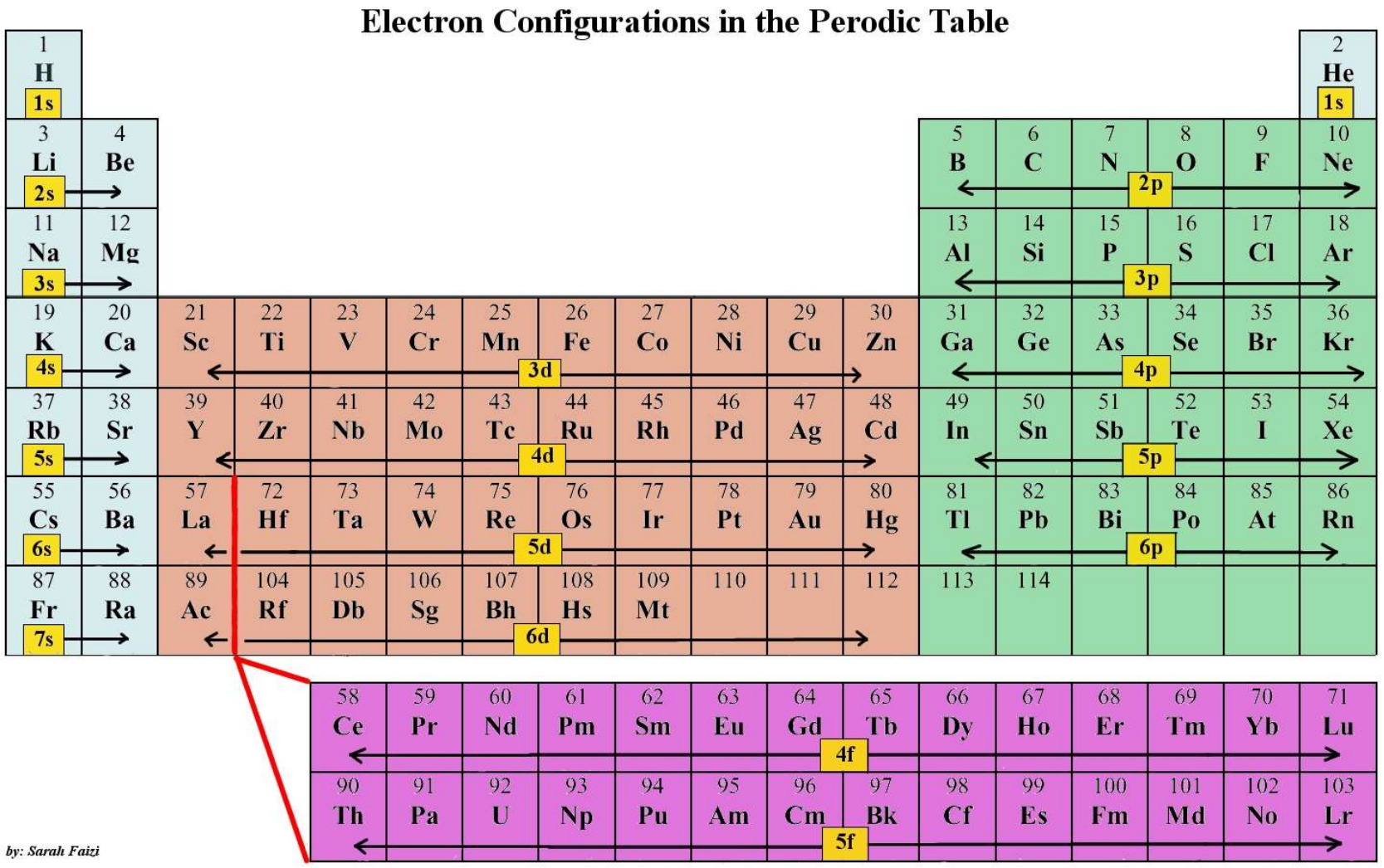


Electrons fill the orbitals in order of increasing energy:



https://en.wikibooks.org/wiki/High_School_Chemistry/Electron_Configurations

The arrangement of elements in the Periodic Table follows the electron configurations. Yellow boxes e.g. 2p below refer to the orbitals that are being filled. Orbitals with lower energy levels are assumed to be full.



http://chemwiki.ucdavis.edu/Core/Inorganic_Chemistry/Electronic_Structure_of_Atoms_and_Molecules/Electronic_Configurations

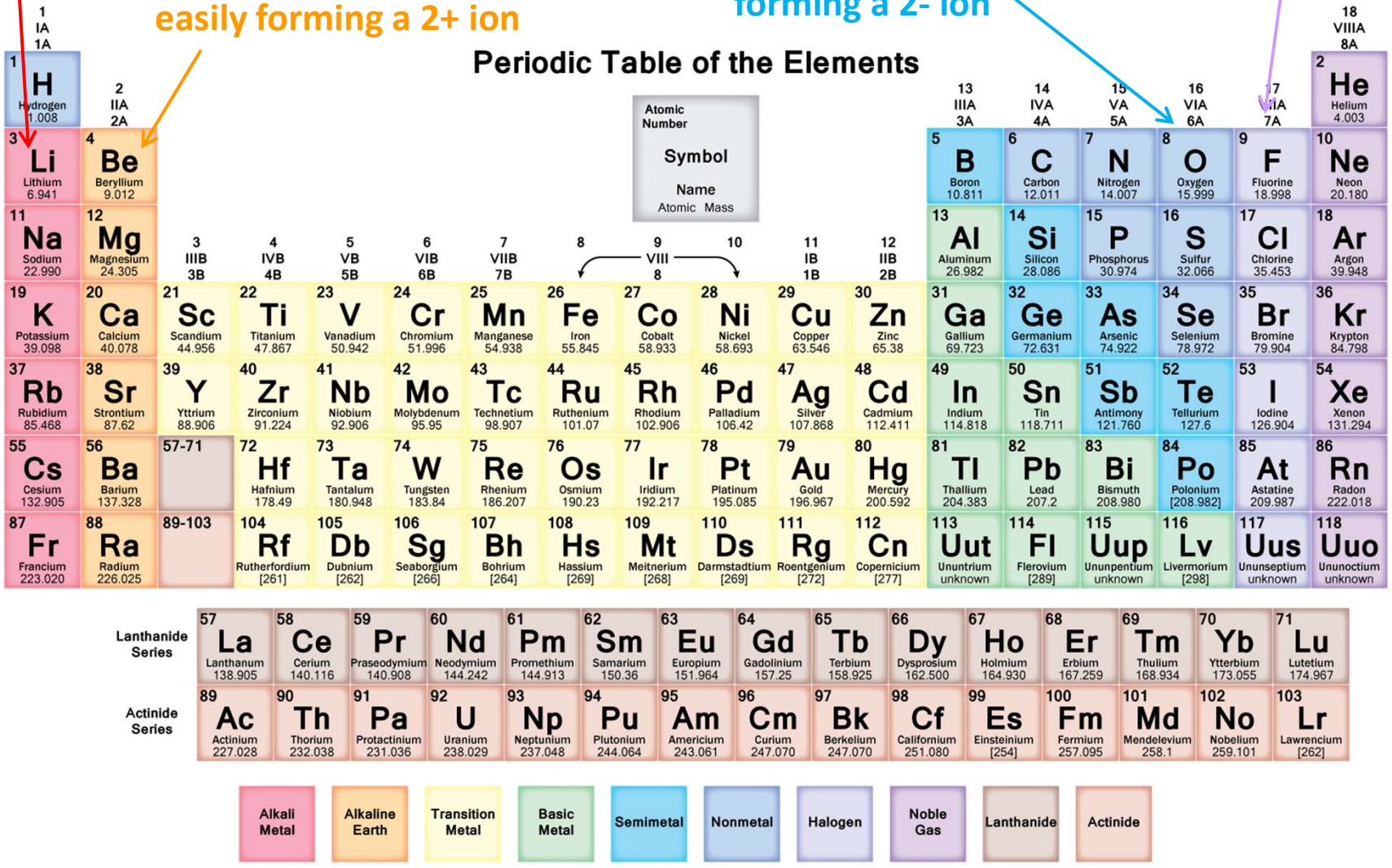
Ionic bonding - elements form ions by losing or gaining electrons

s^1 – 1 electron is lost easily forming a 1+ ion

s^2 – 2 electrons are lost easily forming a 2+ ion

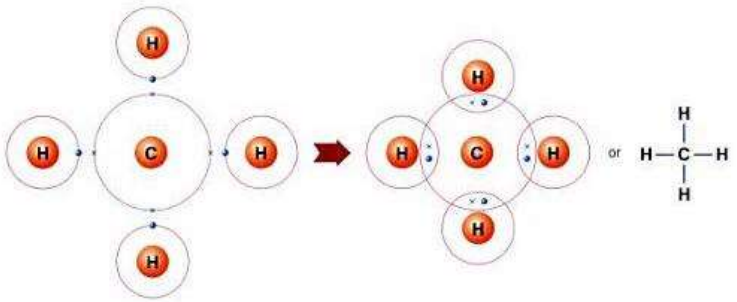
p^4 – 2 electrons can be added forming a 2- ion

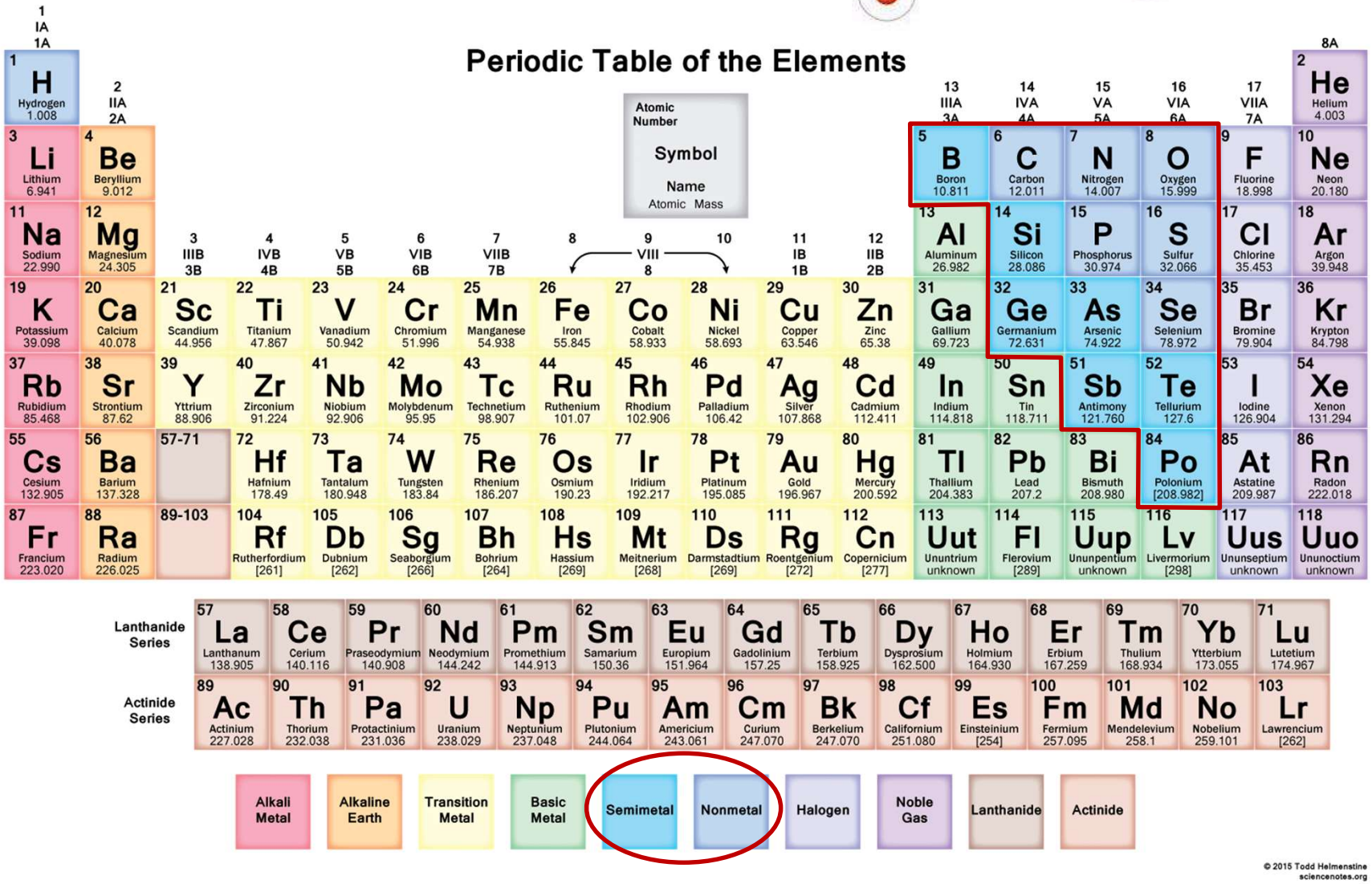
p^5 – 1 electron is easily added forming a 1- ion



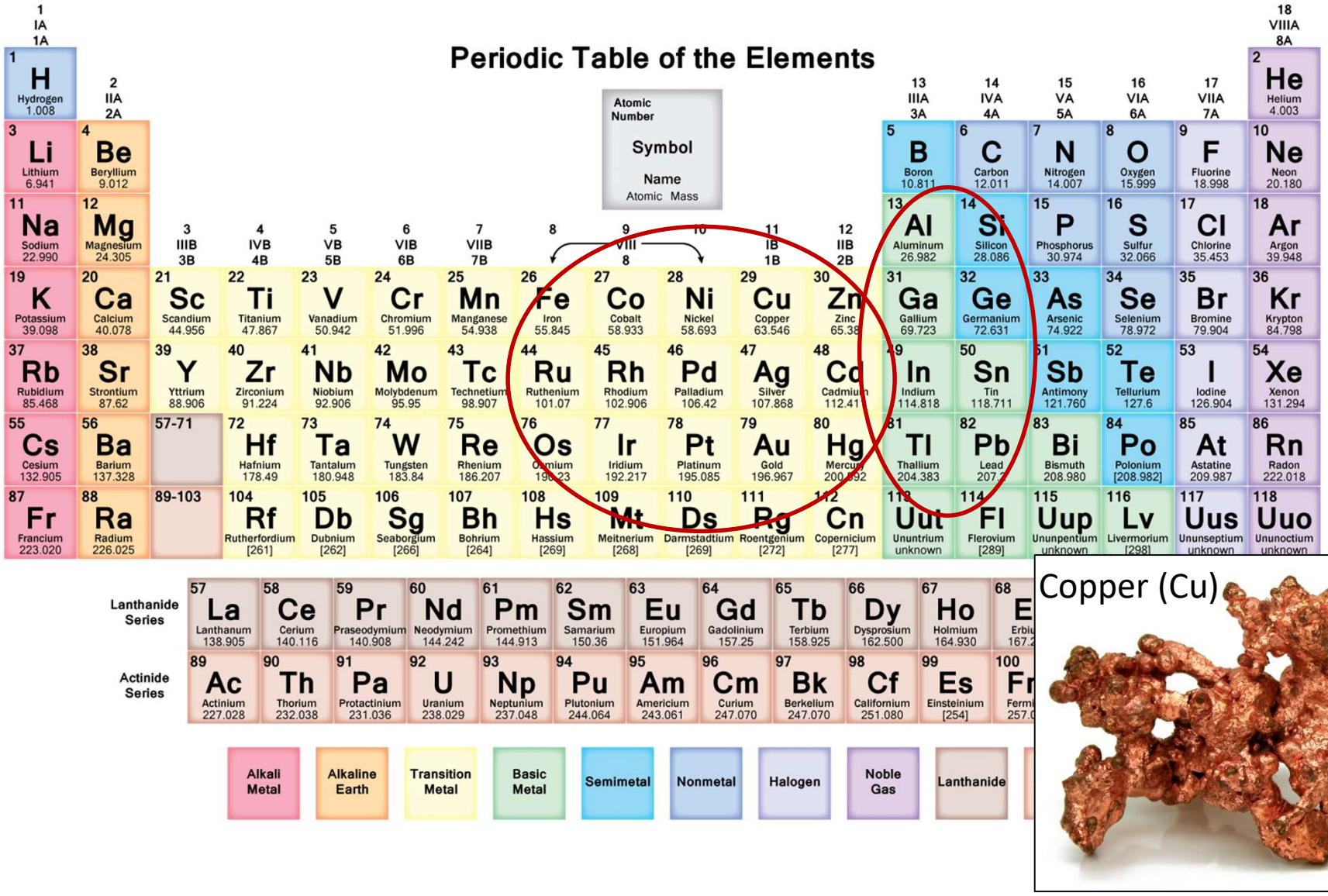
Ionic bonds also in Transition Metals, Lanthanides and Actinides

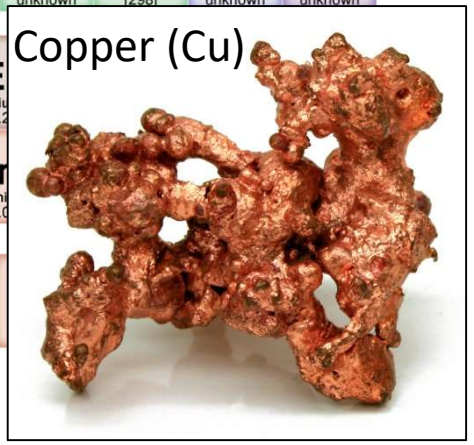
Covalent bonding - atoms share electrons.
No net change in charge.





Metallic bonding – loosely bound cloud of shared electrons.
Common for elements with big outer shells and unfilled d orbitals.

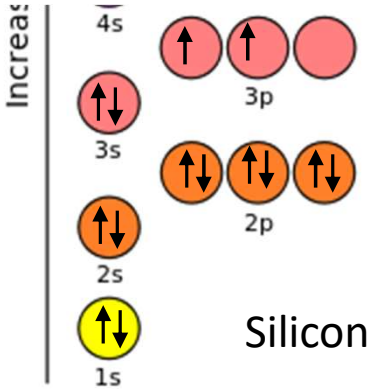




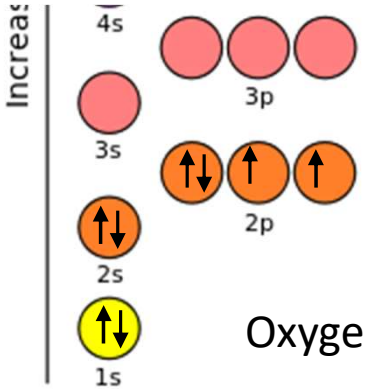
Many of the Transition Metals can form ionic bonds as well as metallic bonds

COMPLEX IONS

Silicate ion – SiO_4 tetrahedron – the building block of silicate minerals



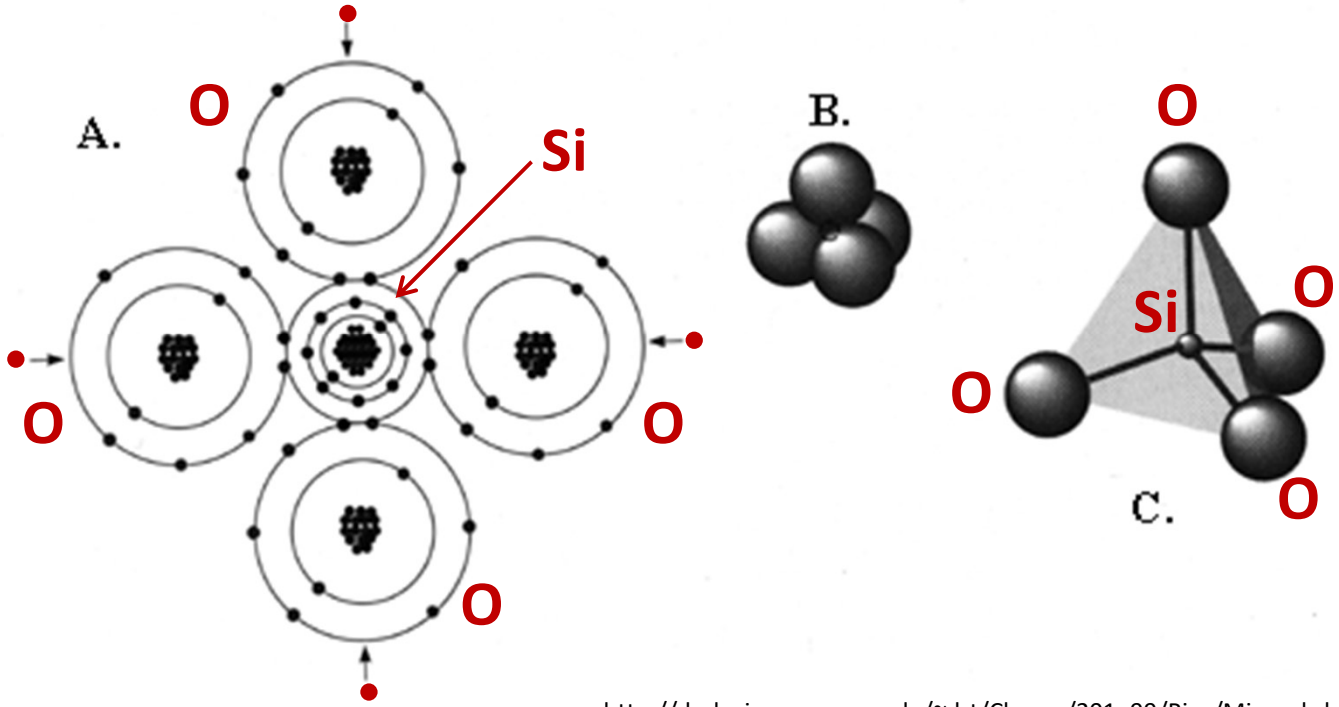
Silicon (Z=14)



Oxygen (Z=8)

Each oxygen atom also adds one extra electron to complete its outer shell.

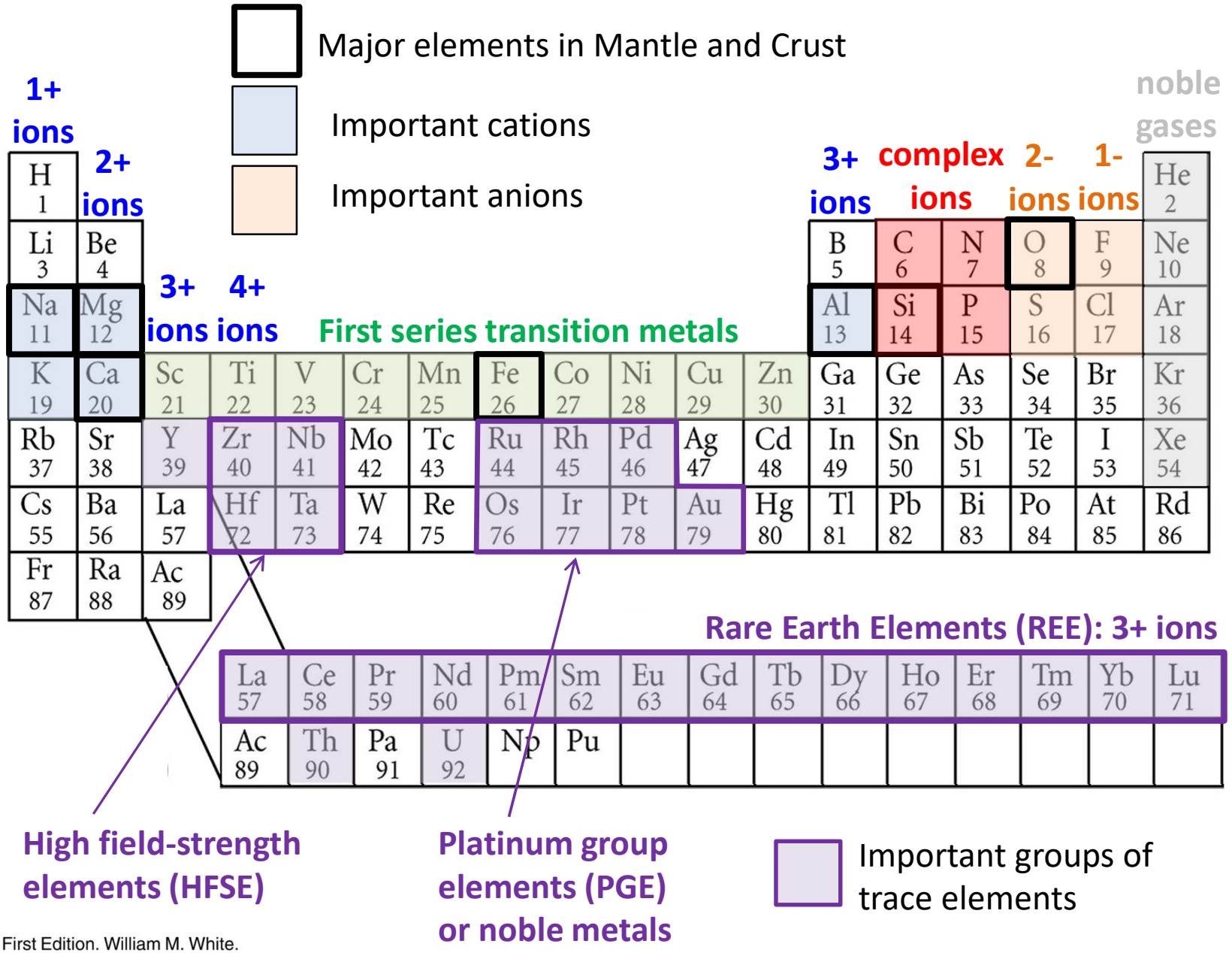
Overall the unit is $[SiO_4]^{4-}$



http://darkwing.uoregon.edu/~drt/Classes/201_99/Rice/Minerals.html

Geochemist's Periodic Table

See Lecture 2



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Chemistry of the Earth

Lecture 3:

Earth's layers and element partitioning

OBJECTIVES OF LECTURE

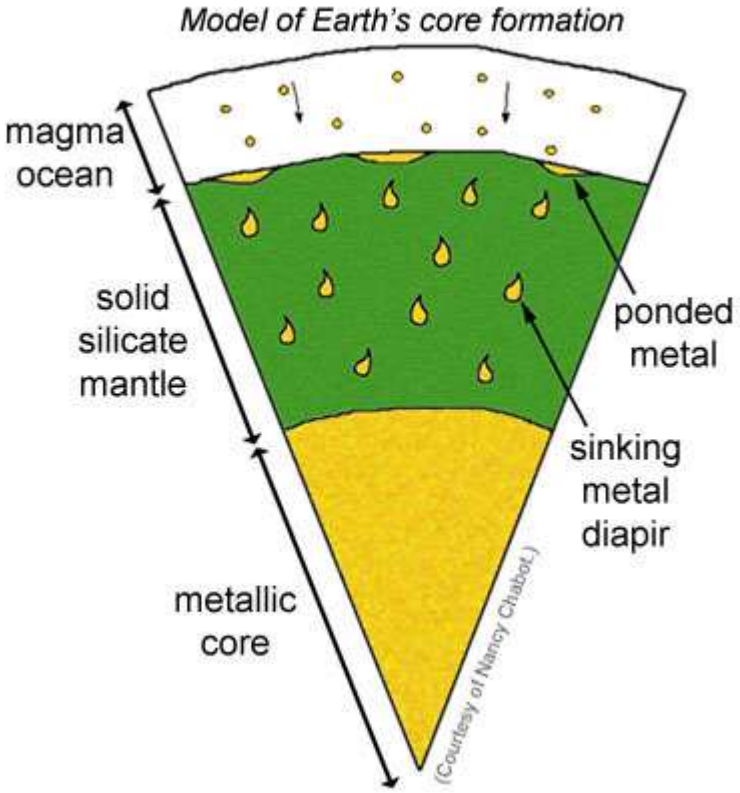
- Understand models for how the Earth's core formed
- Know why chemical elements are classified as siderophile, chalcophile, lithophile, atmophile (Goldschmidt's classification)
- Introduce the concept of element partitioning
- Investigate how the platinum group elements (PGEs) can be used to investigate geologic processes

OBJECTIVES OF PRACTICAL

- Understand the concept of a partition coefficient and practice calculations.
- Investigate one of the hypotheses to explain the high mantle abundances of highly siderophile elements.
- Investigate the hypothesis that a large meteorite impact took place at the K/Pg boundary, using information from highly siderophile elements.

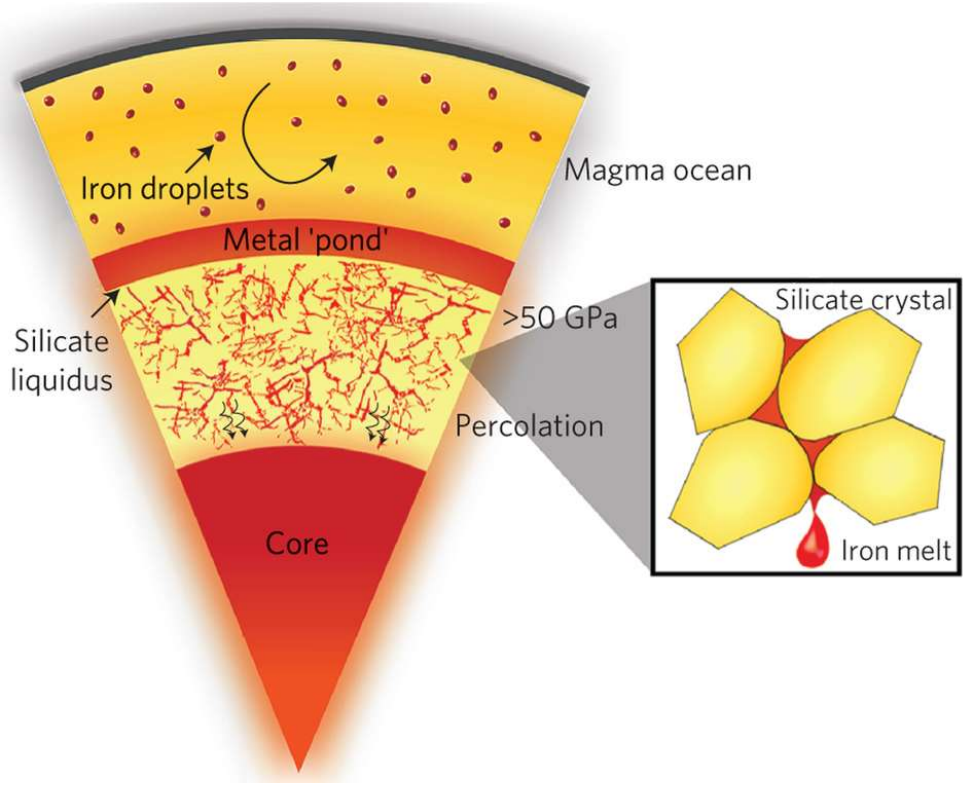
MAGMA OCEAN ON THE EARLY EARTH

Models for core formation include a “magma ocean” layer of molten silicate material.



http://www.psrd.hawaii.edu/July05/cobalt_and_nickel.html

This model has diapirs (pods) of metal sinking through the solid mantle

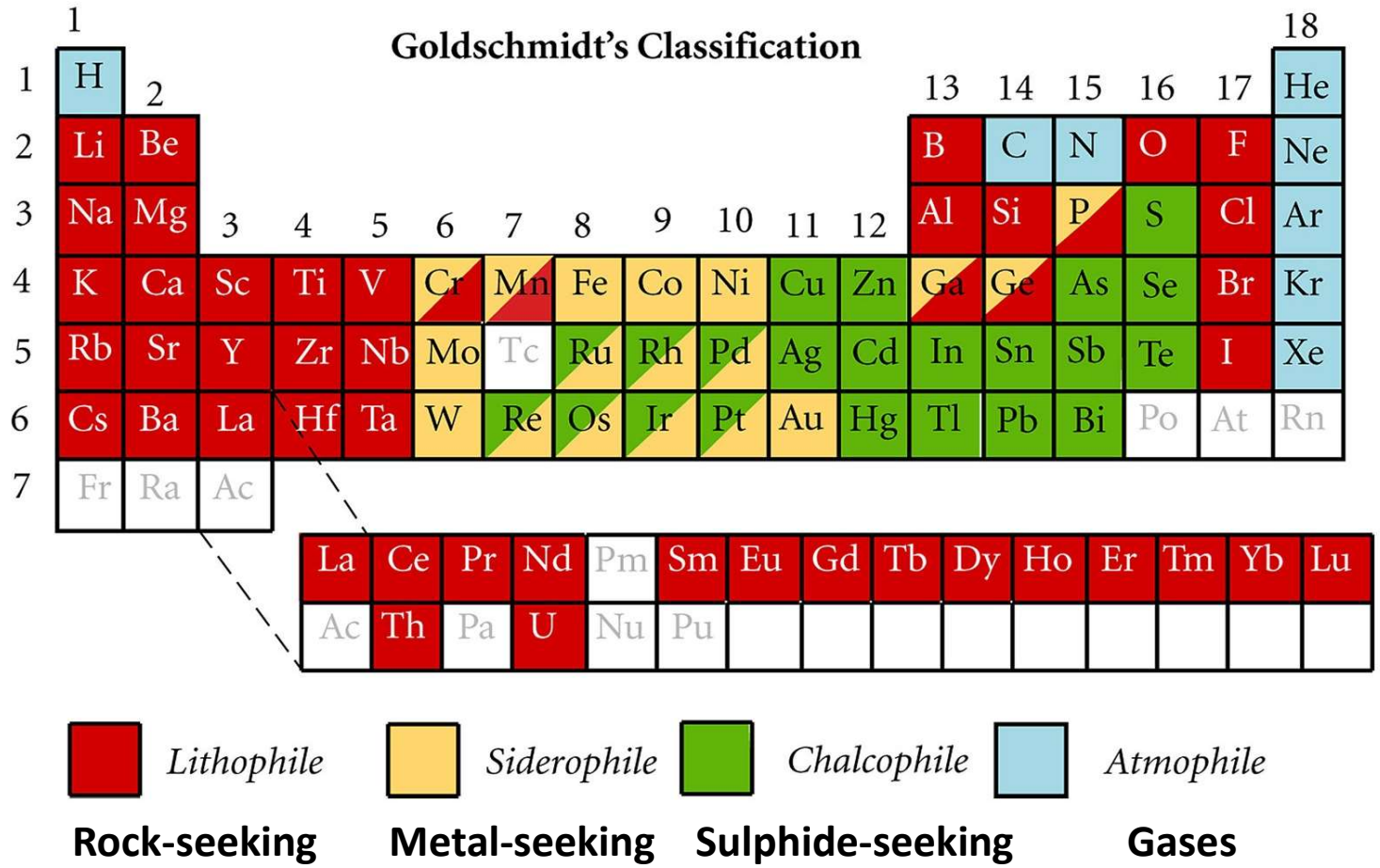


http://www.nature.com/ngeo/journal/v6/n11/pdf/ngeo1956.pdf

This model has liquid metal percolating down through spaces between silicate grains

- Measurements of isotopes such as ^{182}W (tungsten) show that the Earth’s core formed about 50 million years after formation of the Solar System

GOLDSCHMIDT'S CLASSIFICATION OF THE ELEMENTS



- Classification is based on the preference of the element for a silicate, metal or sulphide *liquid*. All three can coexist as immiscible liquids.
- Elements with two colours can occur in either of the two liquids.
- Chemical behaviour is related to the way an element bonds (electron configuration).

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PREFERENCE FOR ONE LIQUID – PARTITIONING

- Preference of an element for one of the liquids is called the *partitioning* behaviour of that element.
- For each element, the degree of preference between two liquids depends on the conditions - such as temperature, pressure, and amount of oxygen available.

We can measure the preference in terms of a ***partition coefficient***:

$$\text{Partition coefficient} = \frac{\text{Abundance in liquid 1}}{\text{Abundance in liquid 2}}$$

Abundances are measured as weight percent (wt%) values, or ppm, ppb etc.

Example:

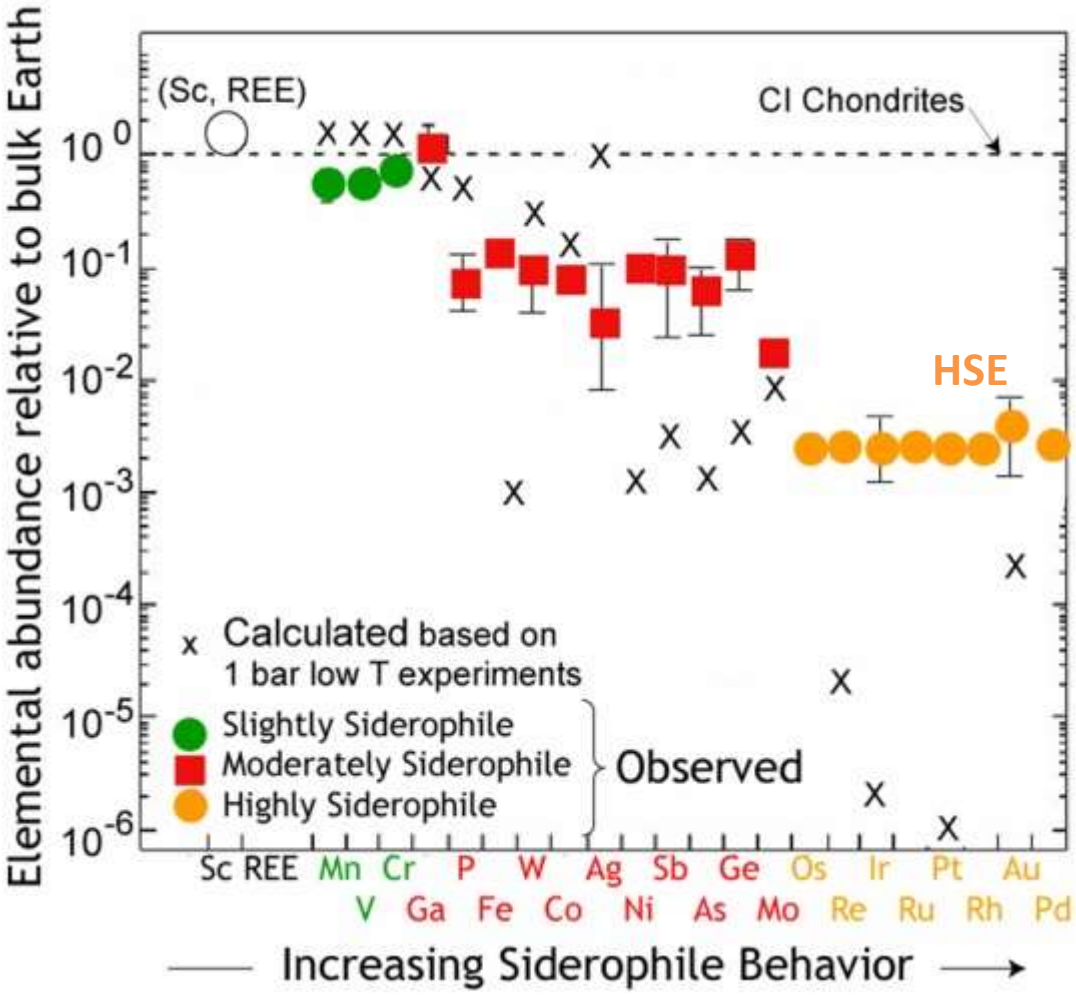
$$\text{Metal / silicate partition coefficient for nickel} = \frac{\text{wt\% Ni in metal}}{\text{wt\% Ni in silicate}}$$

Partition coefficients (also known as distribution coefficients) are given the symbol D.

The example above is: $^{Ni}D_{met/sil}$

MANTLE ABUNDANCES OF HSE ARE ACTUALLY HIGHER THAN EXPECTED

Various hypotheses have been proposed to explain the observations



- Core separation was incomplete and some HSEs were trapped in the mantle.
- Partition coefficients measured at one bar (one atmosphere) do not apply to the high pressures and temperatures of the deep mantle.
- There was late accretion of chondritic material to the Earth, after core formation had taken place.
- More than one of the above processes.

(see Walker 2009, Chemie der Erde)

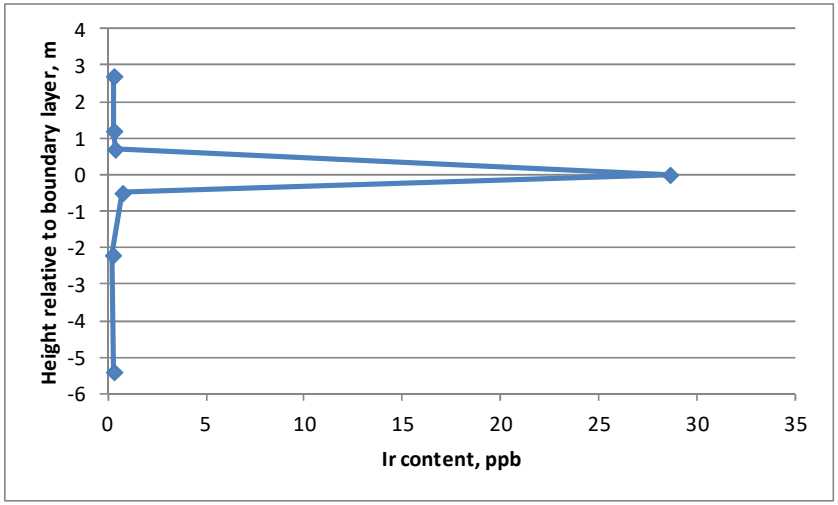
Cottrell 2004: https://www.geol.umd.edu/home/richardwalker/research/siderophile.php

THE DEMISE OF THE DINOSAURS

The mass extinction of many species, including dinosaurs, at the end of the Cretaceous period, has been attributed to the impact of a large meteor.

Highly siderophile elements play a role in supporting this hypothesis.

Plot from practical:





NASA



Getty images

Chemistry of the Earth

Lecture 4:

Global Carbon Cycle and Organic Chemistry

OBJECTIVES OF LECTURE

- Introduction to the concept of (bio)geochemical cycles, in particular the global Carbon Cycle
- Review the basics of organic chemistry
- Understand the importance of organic material in the Carbon Cycle

OBJECTIVES OF PRACTICAL

- Familiarization with the notation of organic chemistry
- Investigation of organic reactions in the petroleum industry, this being related to the anthropogenic (human) contribution to the Carbon Cycle

Geochemical Cycles (Biogeochemical Cycles)

Interactions between:

- Geosphere – the solid Earth (rocks, minerals, soils)
- Hydrosphere – water (oceans, rivers, lakes)
- Biosphere – living organisms
- Atmosphere – gases

Considerations of

- Global reservoirs
- Transfers between reservoirs

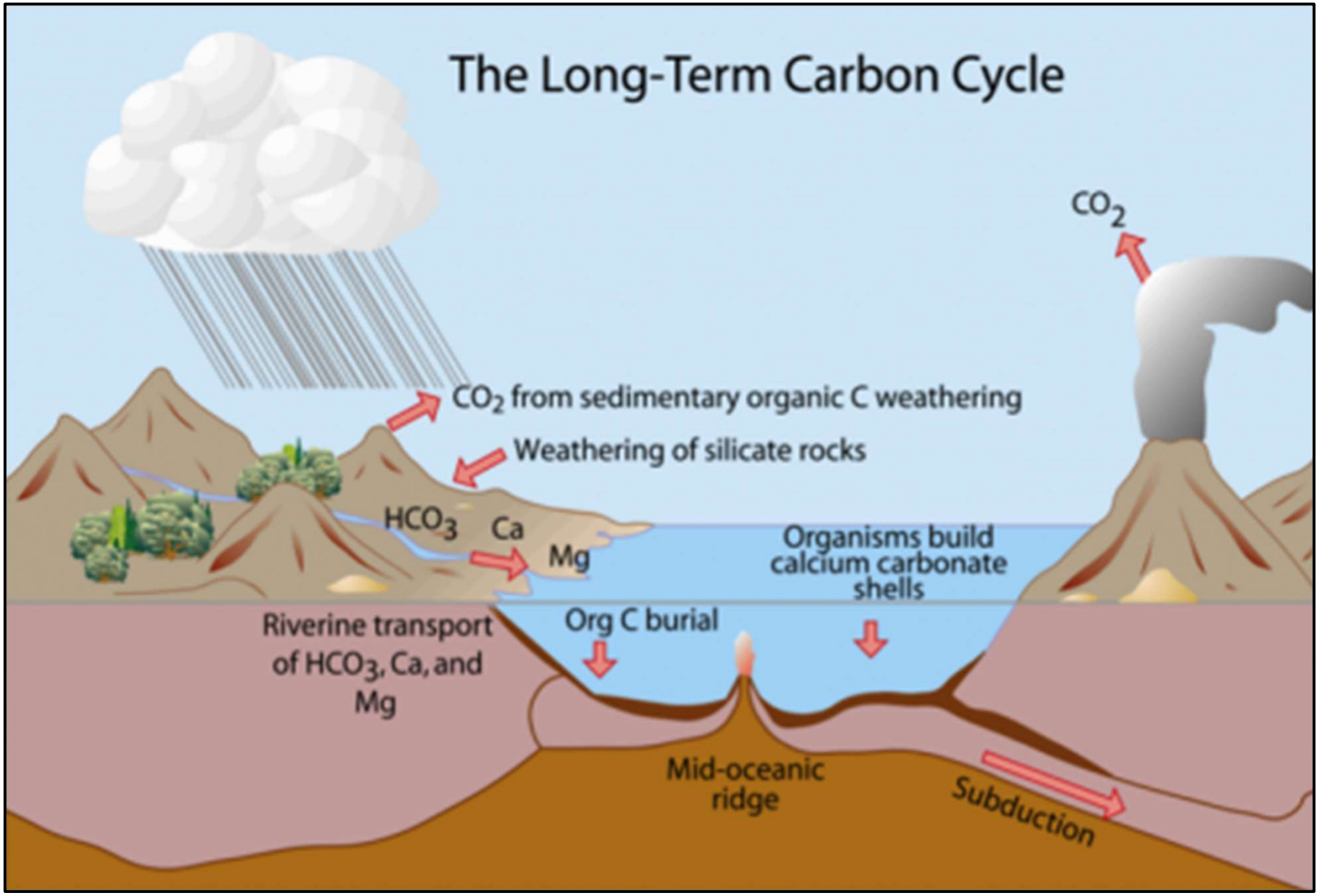
Important cycles:

Carbon, Oxygen, Nitrogen, Sulphur, Phosphorus

Tectonics links two components of the cycles:

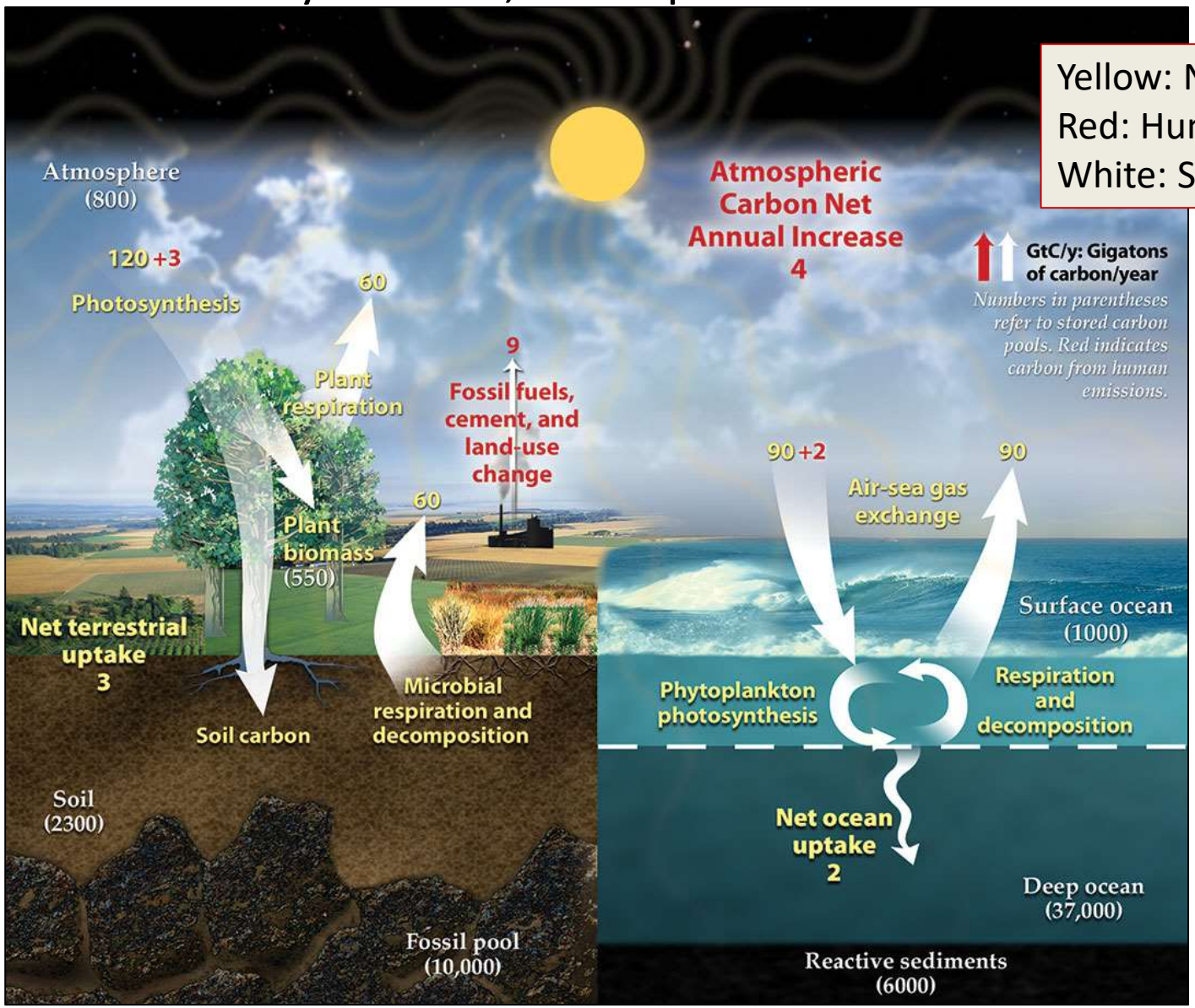
- Exogenic component – operates on Earth's surface
- Endogenic component – operates in Earth's interior

The Carbon Cycle



https://uwaterloo.ca/wat-on-earth/news/microbes-minerals-and-isotopes

Fast carbon cycle: land, atmosphere and oceans



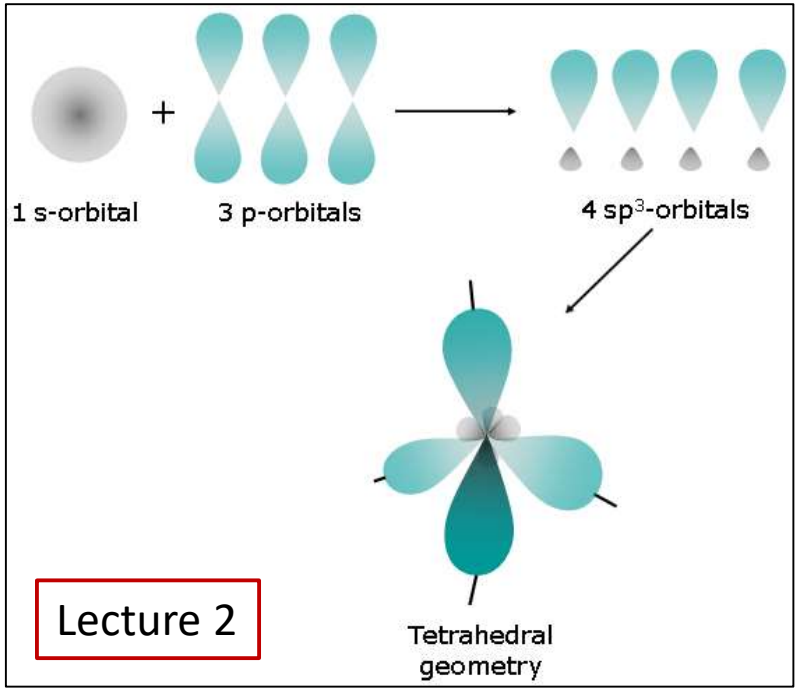
Yellow: Natural fluxes
Red: Human contributions
White: Stored carbon, Gt

The cycle would be balanced without human intervention

Figure shows movement of carbon between land, atmosphere and oceans in billions of tons per year (GtC/y).

http://www.carboncycle.org/the-carbon-cycle/

ORGANIC CHEMISTRY: The Chemistry of Carbon

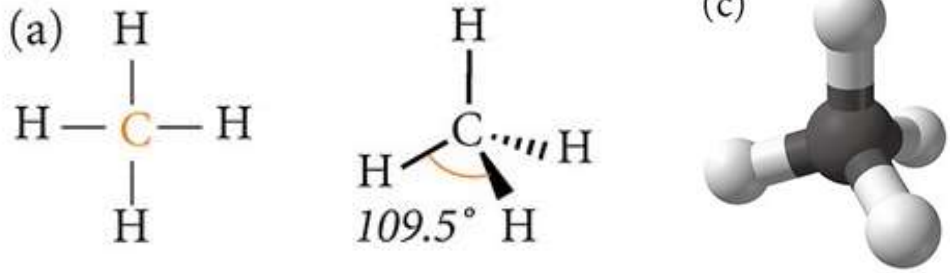


Lecture 2

https://jahschem.wikispaces.com/electronic+structure

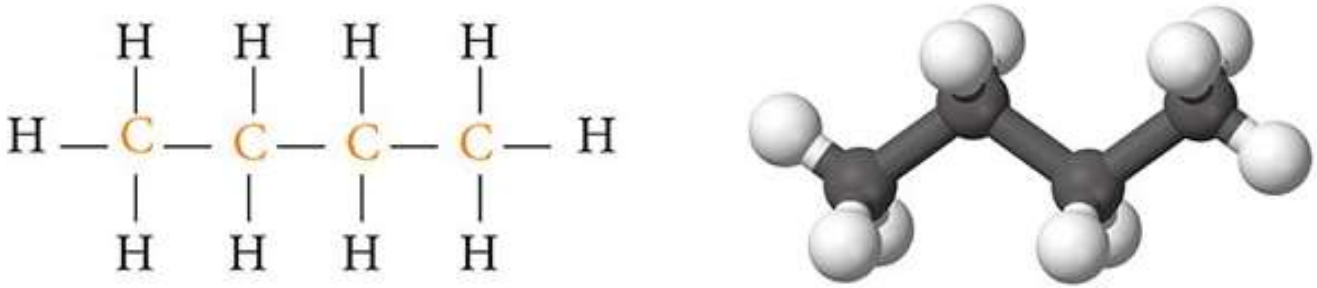
Chain hydrocarbons (Alkanes)

- contain only C and H



Methane (CH_4)

• All bonds are covalent – electrons are shared



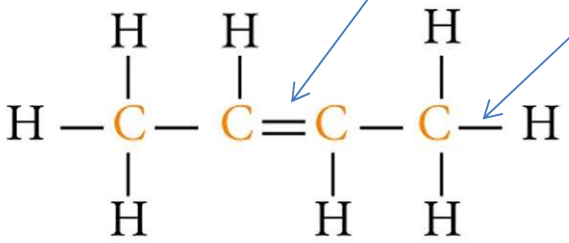
Butane (C_4H_{10})

White Fig. 12.2

Double and triple bonds (Alkenes, Alkynes)

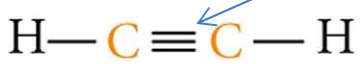
Double bond: 4 electrons shared

Single bond: 2 electrons shared



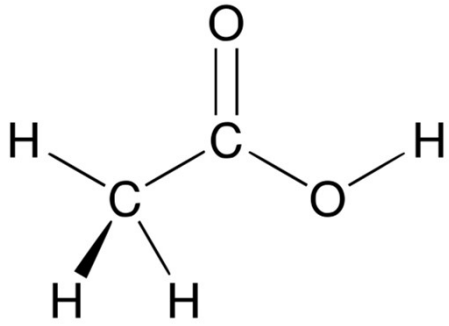
2-Butene

Triple bond: 6 electrons shared



Acetylene

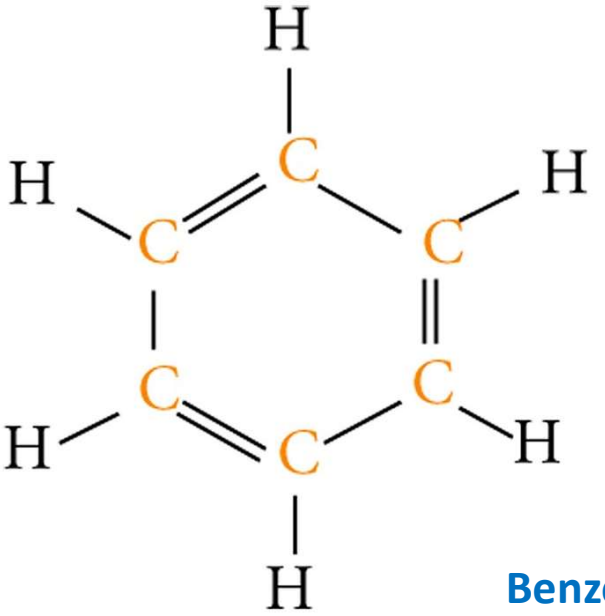
Acetic acid
(Acid or carboxyl group is C=OOH)



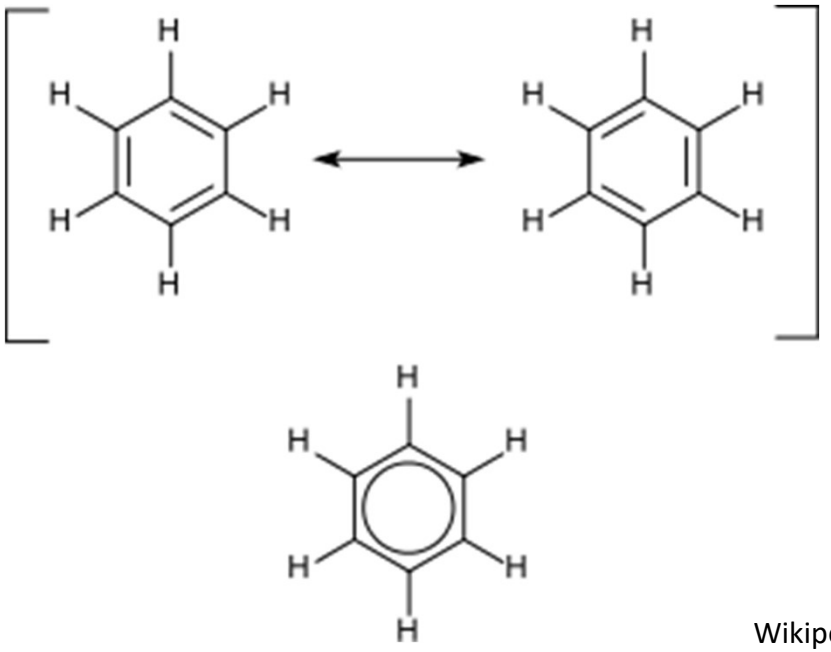
Organic compounds that have chain structures are called **ALIPHATIC.**
Add-ons like –COOH, $–CH_3$ and –OH are called **FUNCTIONAL GROUPS.**

White Fig. 12.4

Ring Structures – Aromatic compounds



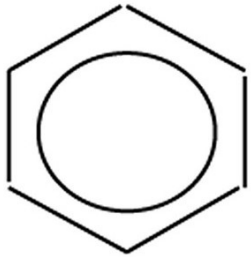
Benzene (C_6H_6)



Wikipedia

The **benzene** ring:

- Six C atoms form a ring.
- Alternating single and double bonds – electrons become delocalized and spread uniformly around the ring.
- The bonding is written as a circle.

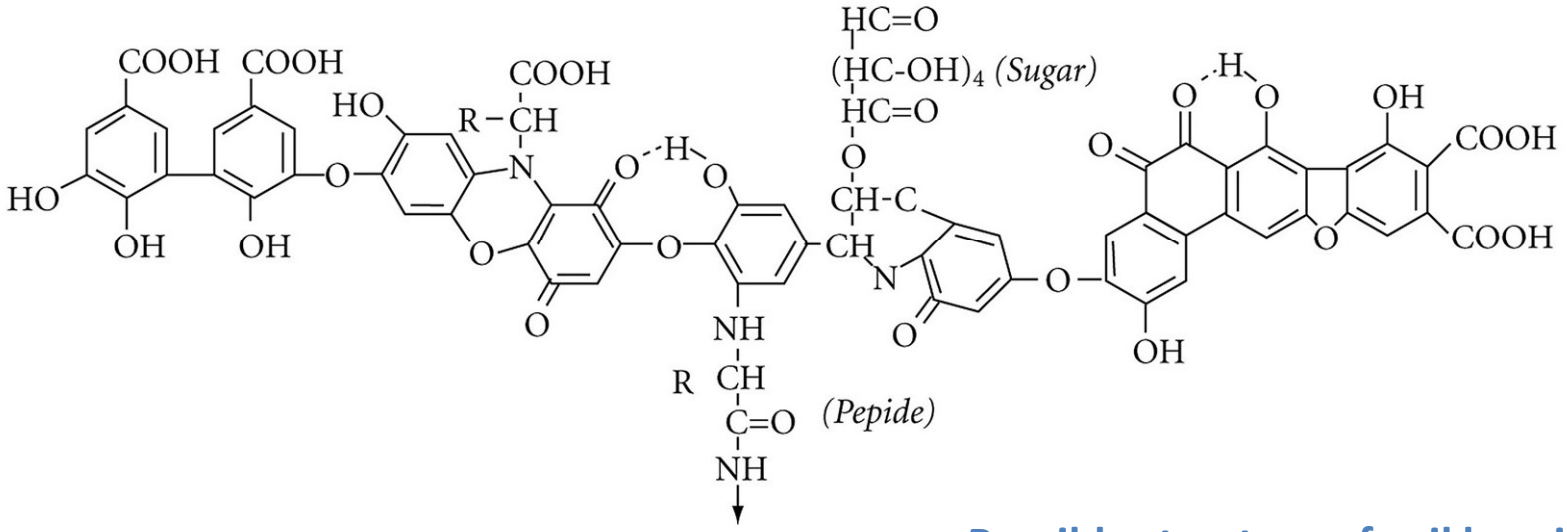


White Fig. 12.5

Organic matter in sedimentary rocks:

Soils: Humic and fulvic acids are produced by biodegradation of organic matter.

Kerogen: a mixture of organic compounds that occur in sedimentary rocks. Formed from humic and fulvic acids. High molecular weight organic compounds (can be >1000 a.m.u.). Most abundant organic material in the Earth's crust.



Possible structure of soil humic acid.
Dashed lines represent hydrogen bonds.

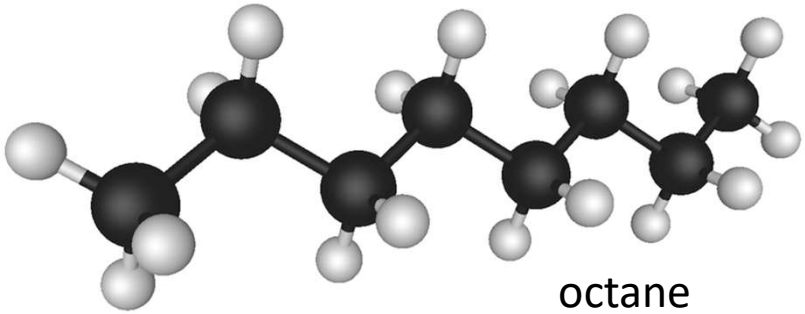
White Fig. 12.23

Organic matter in sedimentary rocks:

Formation of oil and natural gas:

Breakdown of kerogen forms **oil** at 50-150 °C and **natural gas** at 150-200 °C.

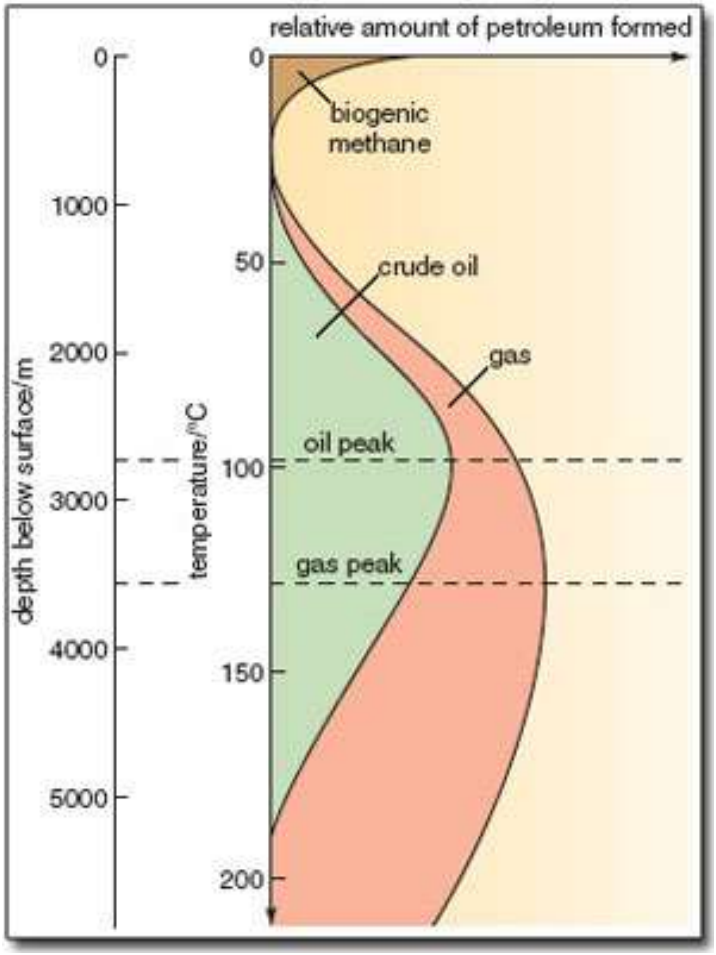
Oil: long chain hydrocarbons



octane

Natural gas: Methane

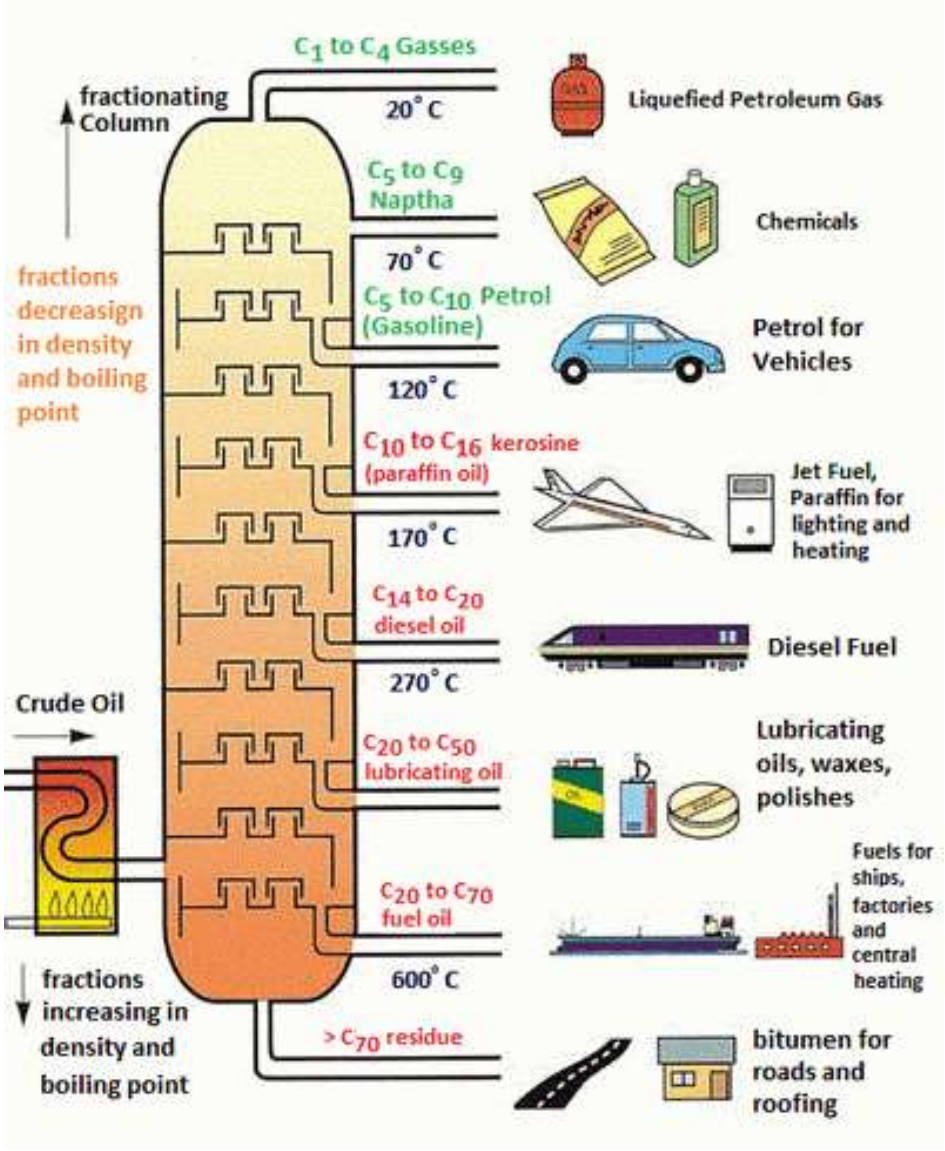




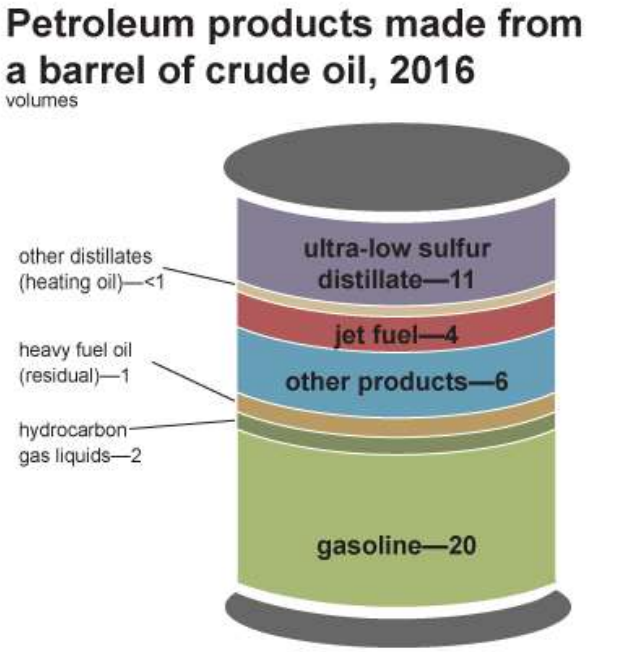
Also see White Fig. 12.37

http://www.open.edu/openlearnworks/mod/page/view.php?id=41003

OIL REFINERY



https://www.erl.com.bd/refiningprocess.php



Chemistry of the Earth

Lecture 5:
Chemical Reactions and Oxidation State

OBJECTIVES OF LECTURE

- Review how chemical formulae and chemical reactions are written, and how to balance a chemical equation
- Review the meaning of moles in chemistry
- Understand the concept of oxidation state, and variable oxidation states among the transition elements
- Investigate some important oxidation-reduction reactions in geochemical systems

OBJECTIVES OF PRACTICAL

- Practice balancing chemical reactions relevant to minerals and rocks
- Practice calculations involving moles and molecular mass
- Investigate some important oxidation-reduction reactions in geochemical systems

Balancing chemical reactions

Reaction:

Olivine + Water → Serpentine + Brucite

2 Mg_2SiO_4 + **3** H_2O → $Mg_3Si_2O_5(OH)_4$ + $Mg(OH)_2$



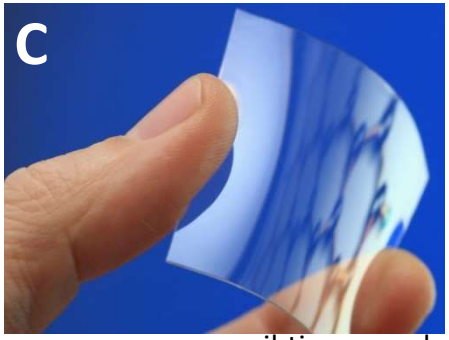
Serpentinite

Element	Olivine **X2**	Water **X3**	Total LHS	=	Serpentine	Brucite	Total RHS
Mg	2 **/ 4**	0	2 **/ 4**		3	1	4
Si	1 **/ 2**	0	1 **/ 2**		2	0	2
O	4 **/ 8**	1 **/ 3**	5 **/ 9 / 11**		9	2	11
H	0	2 **/ 6**	2 **/ 6**		4	2	6

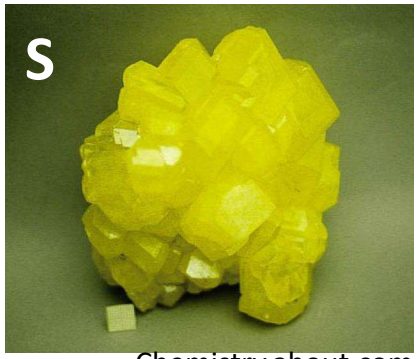
The reaction is now balanced for all elements

The **Avogadro constant** is defined such that the mass of one mole of a substance, expressed in grams, is equal to the atomic or molecular mass of the substance.

12 g of ^{12}C has 6.022×10^{23} carbon atoms
1 g of ^{1}H atoms has 6.022×10^{23} hydrogen atoms
32 g of ^{32}S atoms has 6.022×10^{23} sulphur atoms
.....etc.....



ibtimes.co.uk



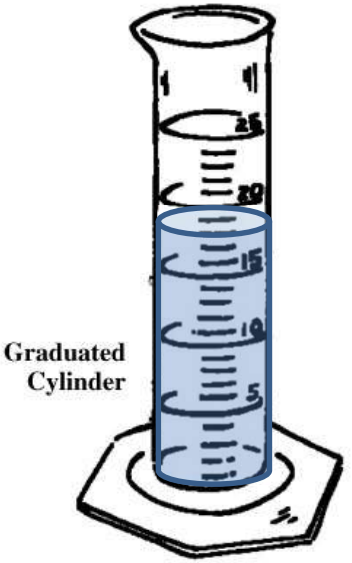
Chemistry.about.com

Water: H_2O

Atomic mass of H is 1
Atomic mass of O is 16

- One molecule of H_2O has a **molecular mass** of 18
- One mole of water has a mass of 18 g

- 18 g of water has 6.022×10^{23} water molecules



OXIDATION – REDUCTION REACTIONS (REDOX REACTIONS)

Lecture 2

Some reactions involve a change in the VALENCE of an element.

Valence – charge on an ion.
Example: Fe^{2+} (divalent iron ion), Fe^{3+} (trivalent iron ion)

Hematite, Fe_2O_3 -- $Fe^{3+}{}_2O^{2-}{}_3$ Magnetite, Fe_3O_4 -- $Fe^{2+}Fe^{3+}{}_2O^{2-}{}_4$

Hematite = Magnetite + O_2
$6\ Fe_2O_3$ = $4\ Fe_3O_4$ + O_2

Valence can also be described in terms of OXIDATION STATE.

Oxidation state of iron is Fe^{2+} (ferrous), Fe^{3+} (ferric).

The above reaction involves a *change of oxidation state*, so it is described as an oxidation-reduction reaction or a REDOX reaction.

OXIDATION – REDUCTION REACTIONS (REDOX REACTIONS)

Hematite = Magnetite + O_2

$6\ Fe^{3+}{}_2O^{2-}{}_3 = 4\ Fe^{2+}Fe^{3+}{}_2O^{2-}{}_4 + O_2$





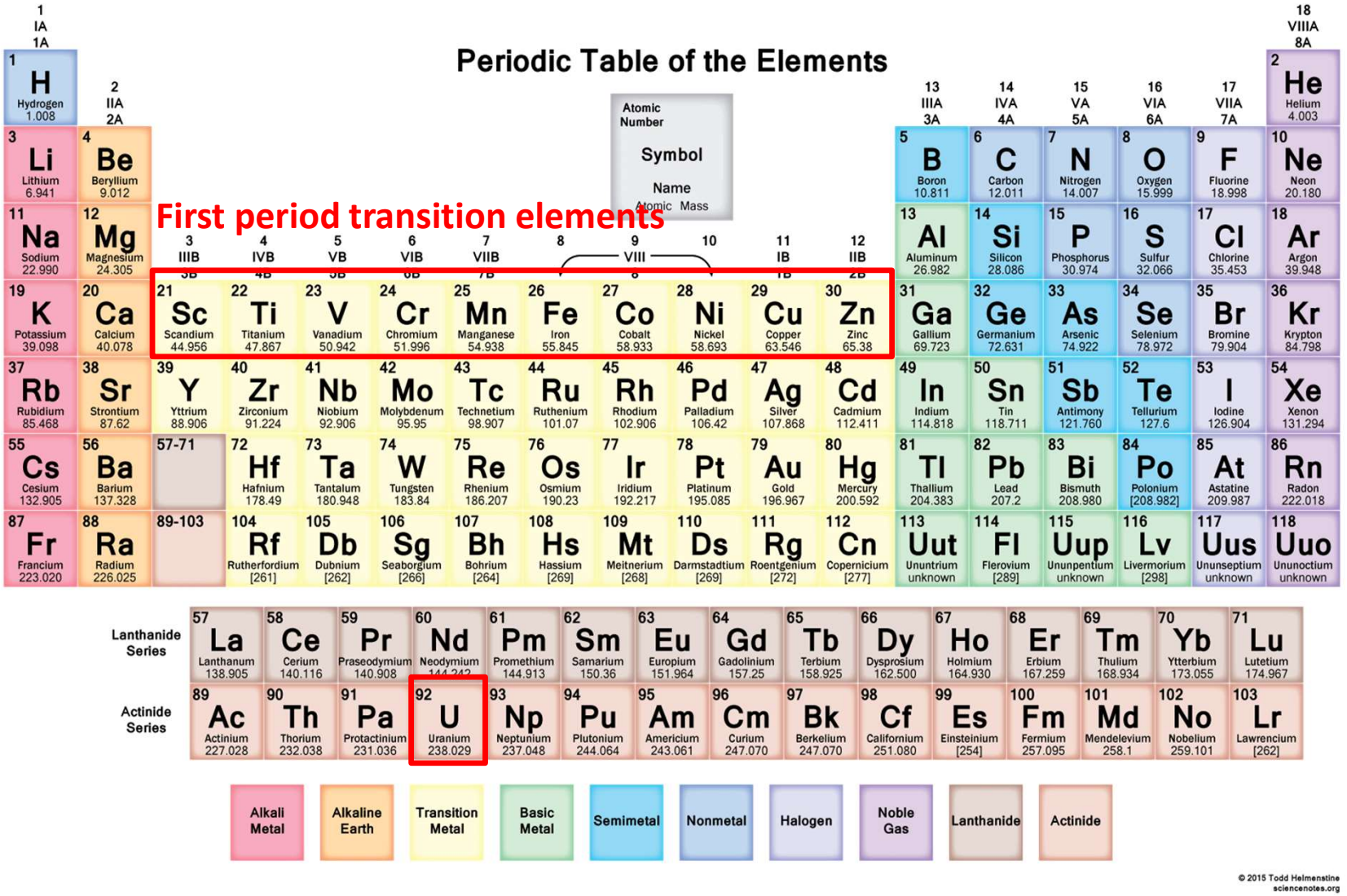
- An Fe^{3+} ion in hematite gains an electron to form Fe^{2+}, and is therefore reduced. Reduction = decrease in oxidation state.

  $Fe^{3+} + e^- = Fe^{2+}$ (reduction of Fe^{3+})

- In the reverse reaction, the Fe^{2+} ion in magnetite loses an electron to form Fe^{3+}, and is therefore oxidized. Oxidation = increase in oxidation state.

  $Fe^{2+} = Fe^{3+} + e^-$ (oxidation of Fe^{2+})

VARIABLE OXIDATION STATE: TRANSITION METALS



Transition elements – filling the d orbitals.

Electron configurations can vary.

Uraninite

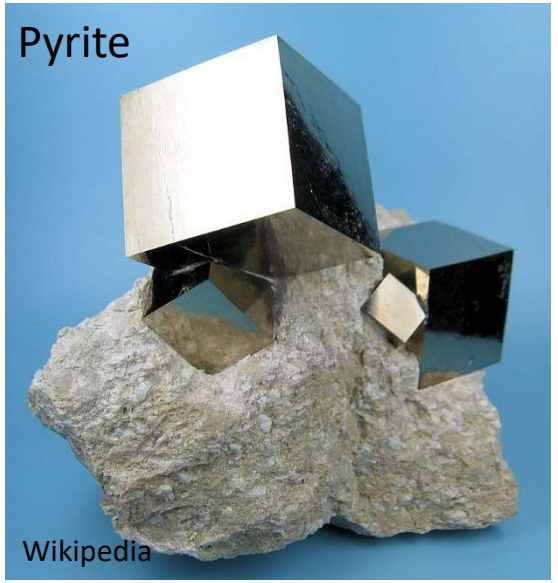


Wikipedia





Pyrite



Wikipedia

Chemistry of the Earth

Lecture 6:
Chemical Equilibrium (Thermodynamics)

OBJECTIVES OF LECTURE

• Understand what is meant by equilibrium in a chemical reaction
• A basic introduction to thermodynamics
• Understand how equilibrium is represented in a phase diagram
• Understand how geochemical systems respond to changes in temperature and pressure (mineral polymorphs, melting)

OBJECTIVES OF PRACTICAL

- Familiarity with thermodynamic variables (enthalpy, entropy, Gibbs Free Energy) and the concept of equilibrium
- Construct a phase diagram from thermodynamic data
- Construct a phase diagram from experimental data
- Interpretation of basic phase diagrams and the Gibbs Phase Rule

COMBINING THE FIRST AND SECOND LAWS

First Law: $dU = dQ - PdV$

Second Law: $dS \geq dQ/T$

Combined: $dU \leq TdS - PdV$

Since most geological process occur very slowly and small changes are close to being reversible, we can assume that:

$$\mathbf{dU = TdS - PdV}$$

FUNDAMENTAL EQUATION of THERMODYNAMICS

U – internal energy
T – temperature in Kelvins
S – entropy
P – pressure
V – volume

GIBBS FREE ENERGY: The amount of energy available for a system to do useful work at constant T and P.

$$G = H - TS$$

For a change in the system,

$$\boxed{\Delta G = \Delta H - T\Delta S}$$

When $\Delta G = 0$, the system is in **chemical equilibrium**.

When $\Delta G < 0$ (i.e. when ΔG is negative), there is energy available to produce a spontaneous change.

A chemical reaction will proceed in the direction of lower Gibbs free energy.

Another useful relationship in thermodynamics:

$$\boxed{dG = VdP - SdT}$$

THERMODYNAMICS IN A NUTSHELL

$PV = nRT$ — **Ideal Gas Law**

$dU = dQ - PdV$ — **First Law of Thermodynamics**

$dS \geq dQ/T$ — **Second Law of Thermodynamics**

$dU = TdS - PdV$

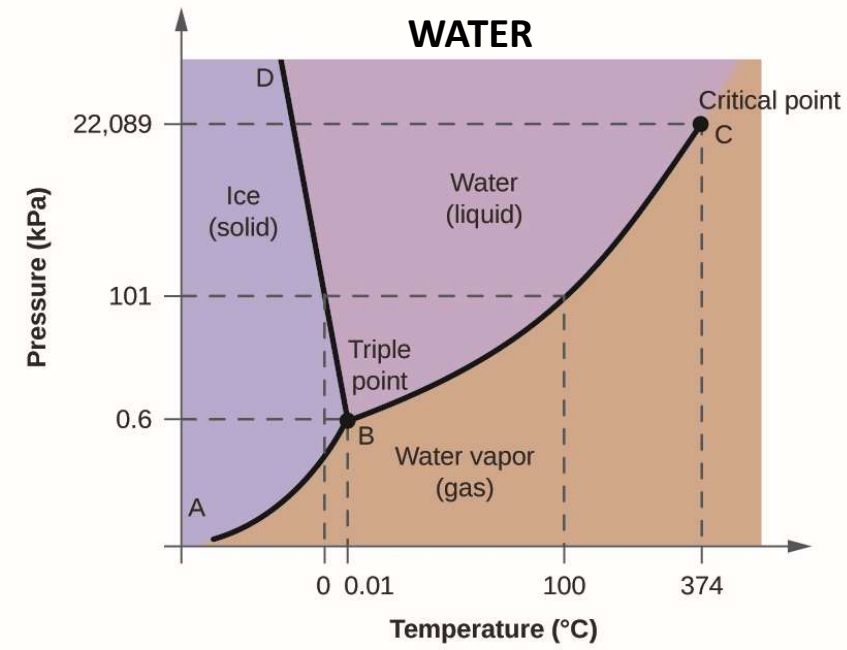
$dG = VdP - SdT$

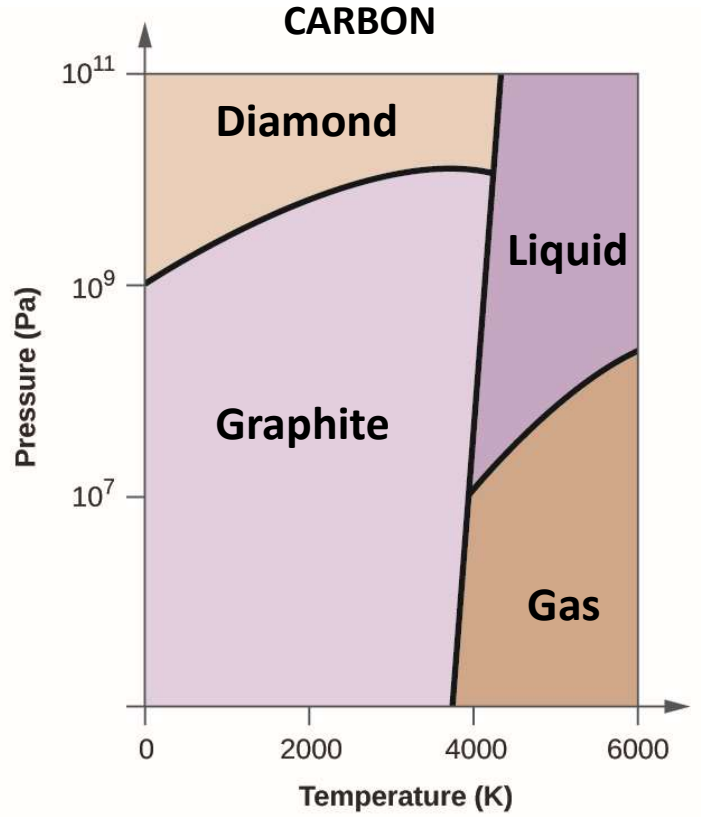
$\Delta G = \Delta H - T\Delta S$

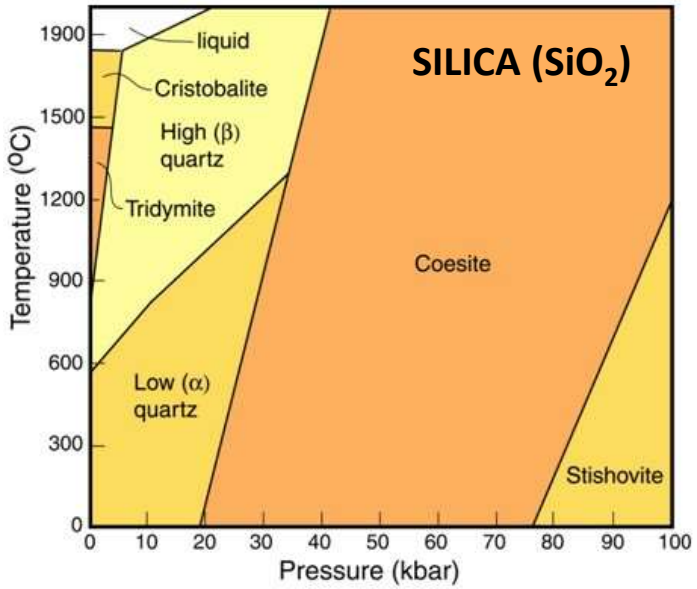
d = a very small change
Δ = a change

P = pressure
V = volume
n = number of moles
R = Gas Constant (8.314 J mol^{-1})
T = temperature **in Kelvin**
U = internal energy (sometimes written as E)
Q = heat flow from surroundings
S = entropy
H = enthalpy
G = Gibbs Free Energy

Some simple one-component phase diagrams:







Axes are temperature and pressure

Silica and Carbon diagrams show the stability fields of mineral polymorphs

GIBBS PHASE RULE:

An equation relating the number of phases, components and thermodynamic variables in a system.

$$\mathbf{f = c - \phi + 2}$$

c = components

ϕ = phases

f = "degrees of freedom"

f – number of intensive variables that can be changed independently, without altering the state of the system, i.e. without changing the number of phases. Also termed "variance".

f – usually consider the variables temperature (T) and pressure (P).

- The "2" in the above equation applies to a case where X is fixed, and we are considering changes in T and P.
- If pressure is fixed e.g. in a T vs X diagram, the phase rule becomes: $\mathbf{f = c - \phi + 1}$

Chemistry of the Earth

Lecture 7:
Melting and
Rare Earth Elements (REE)

OBJECTIVES OF LECTURE

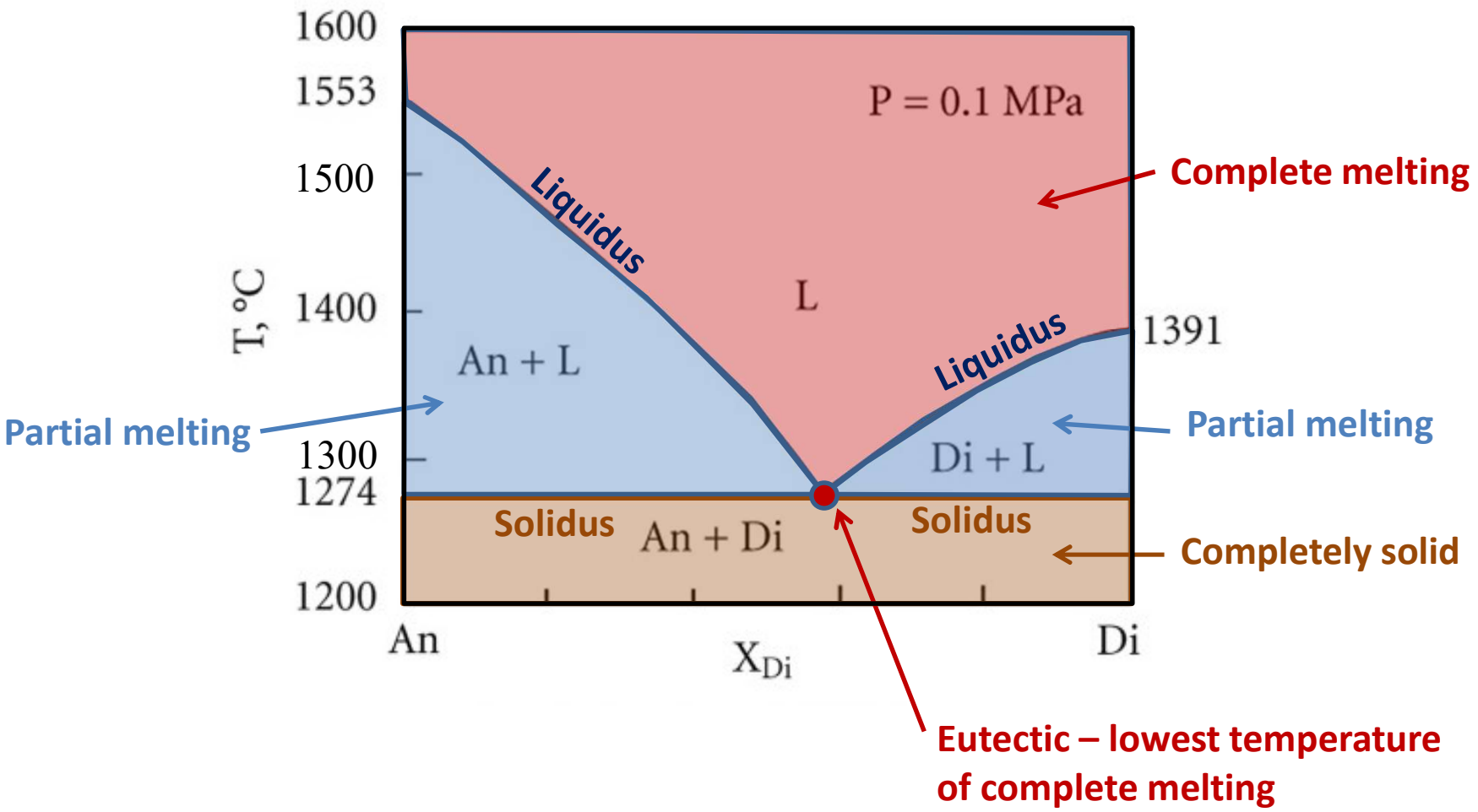
• Investigate how and why rocks melt
• Use phase diagrams to understand how rocks melt and crystallize over a range of temperatures
• Investigate the relationship between rock chemistry and melt chemistry, and the concept of element partitioning
• Understand how and why the rare earth elements act as powerful tracers of geochemical processes

OBJECTIVES OF PRACTICAL

- Learn how to interpret partial melting of a rock as shown in a phase diagram
- Understand why the Rare Earth Elements (REE) are powerful trace elements that can be used to interpret geochemical processes:
    - Investigate the behaviour of REE in different minerals
    - Use the Rare Earth Elements (REE) to interpret melting processes in the mantle

PARTIAL MELTING:

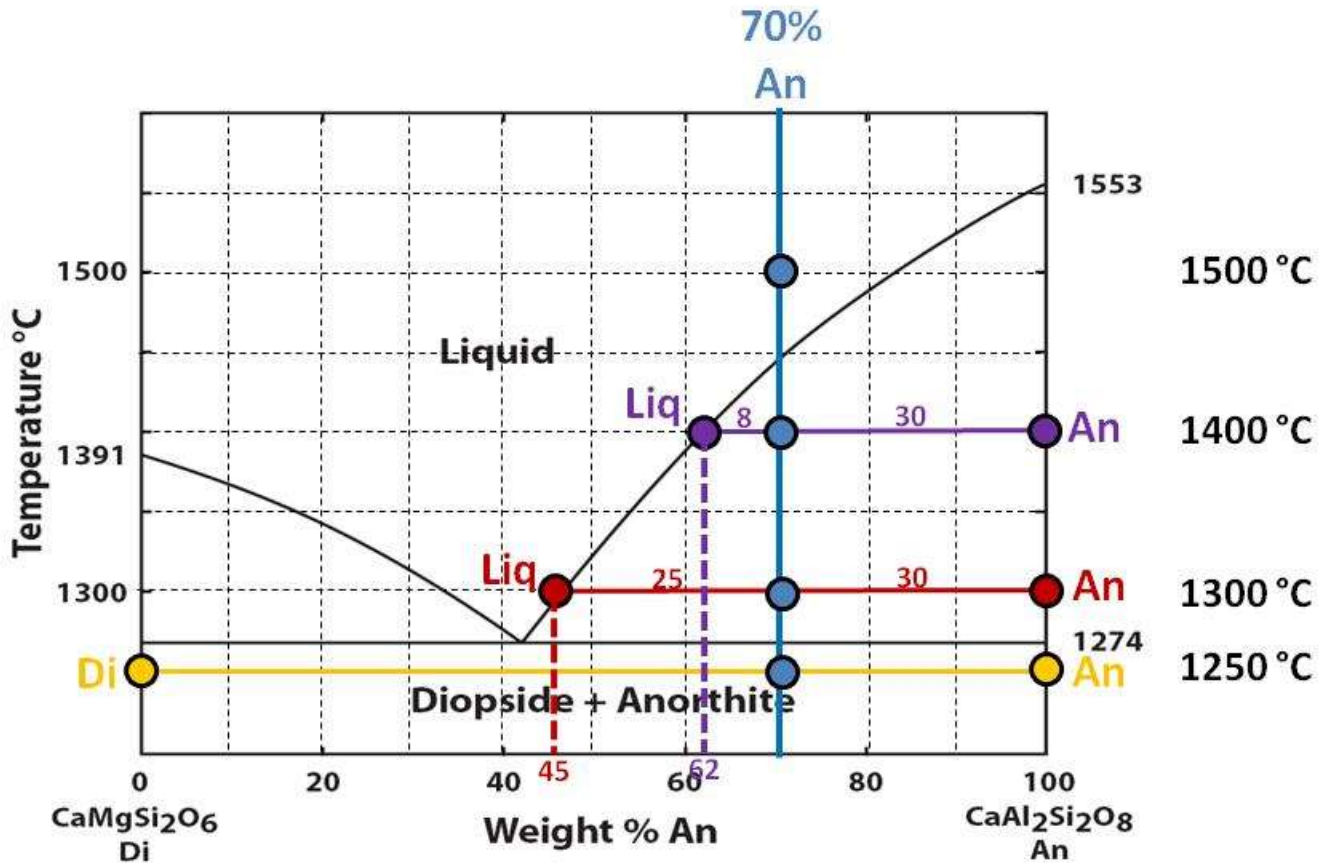
A rock that consists of a mixture of minerals melts over a range of temperatures. When both melt and solid material are present, the rock is partially molten.



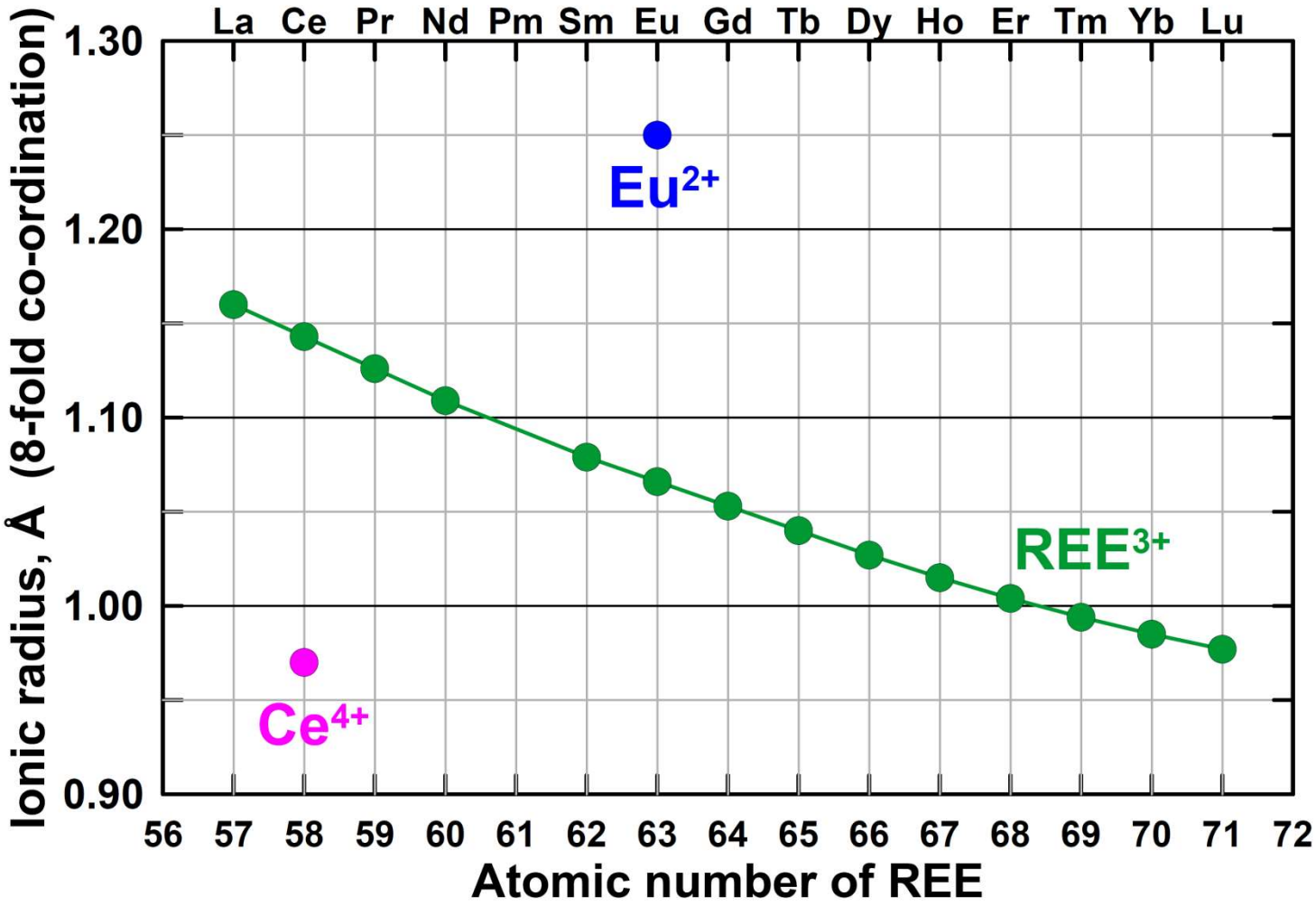
PARTIAL MELTING:

A rock that consists of a mixture of minerals melts over a range of temperatures. When both melt and solid material are present, the rock is partially molten.

Example: A mixture of 70% Anorthite and 30% Diopside (see practical).
Phase diagram shows *proportions* of phases present as well as *compositions* of phases that have variable compositions (in this case, liquid).



REE: Lanthanide contraction



- The ionic radius of REE ions with charge 3+ decreases steadily with increasing Z.
- Reason is that the f orbital electrons do not completely shield the charge of the nucleus. As the charge on the nucleus increases, the electrons are pulled closer in.

PARTITION COEFFICIENTS OF REE IN DIFFERENT MINERALS - OLIVINE, PYROXENE, PLAGIOCLASE, GARNET, SPINEL, AMPHIBOLE

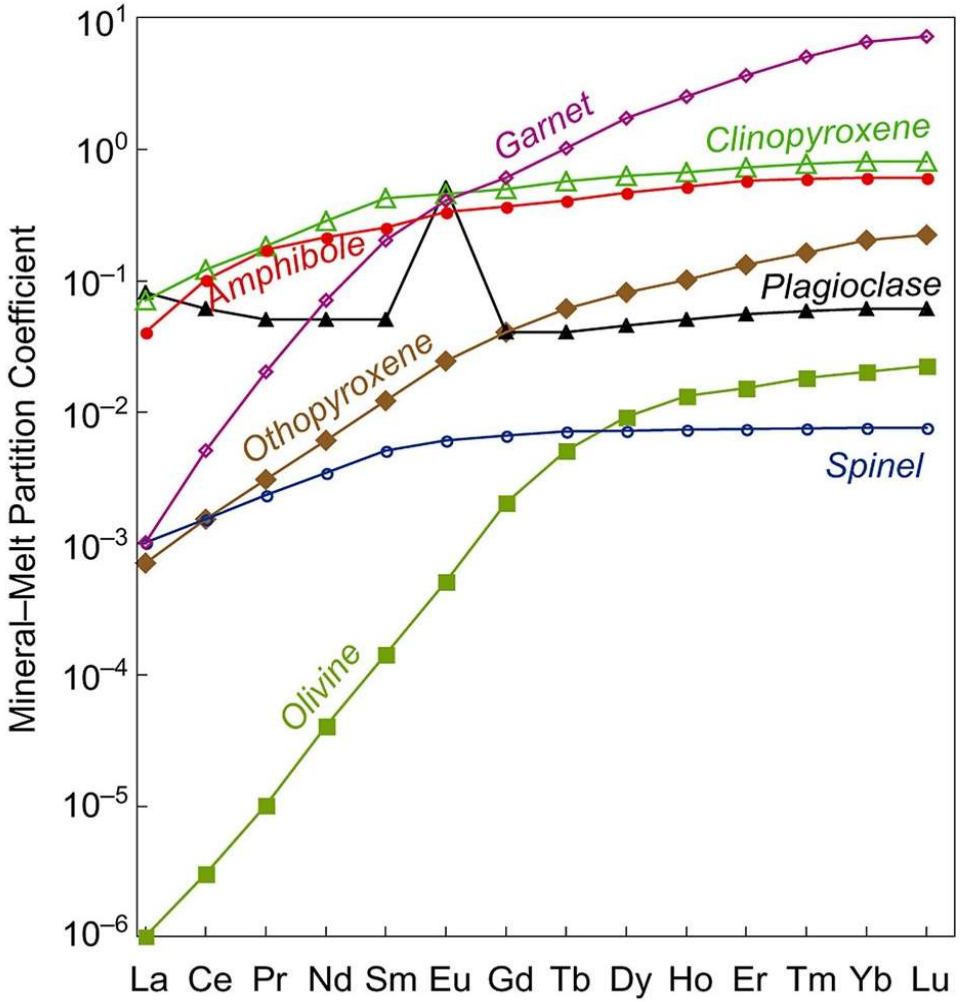
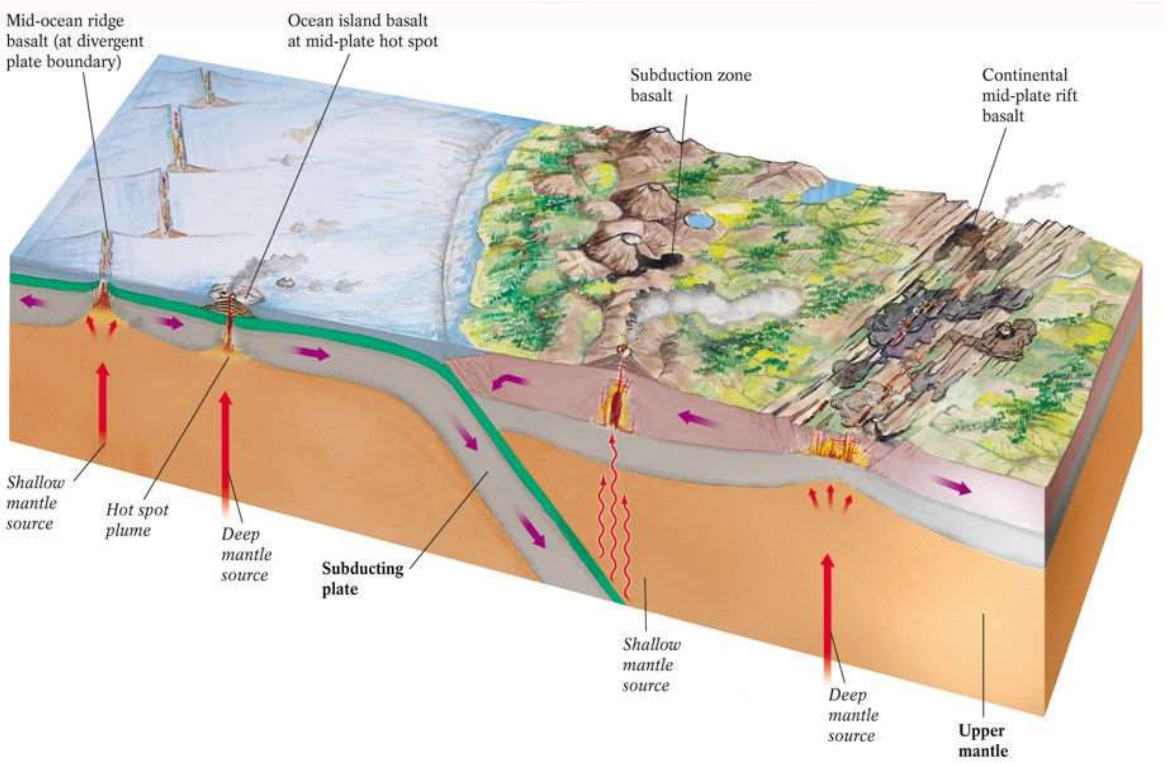


Figure 7.16 Rare earth mineral–melt partition coefficients for mafic magmas. Data from Table 7.5.

White, Geochemistry

- Patterns vary smoothly across the series because of smooth changes in ionic radii.
- Eu is an exception because it is commonly 2+ not 3+.
- In most minerals, all REE are incompatible (D < 1).
- In most minerals, La is more incompatible than Lu.
- Because partition coefficients are a ratio, there is no need to normalize in order to get smooth patterns.

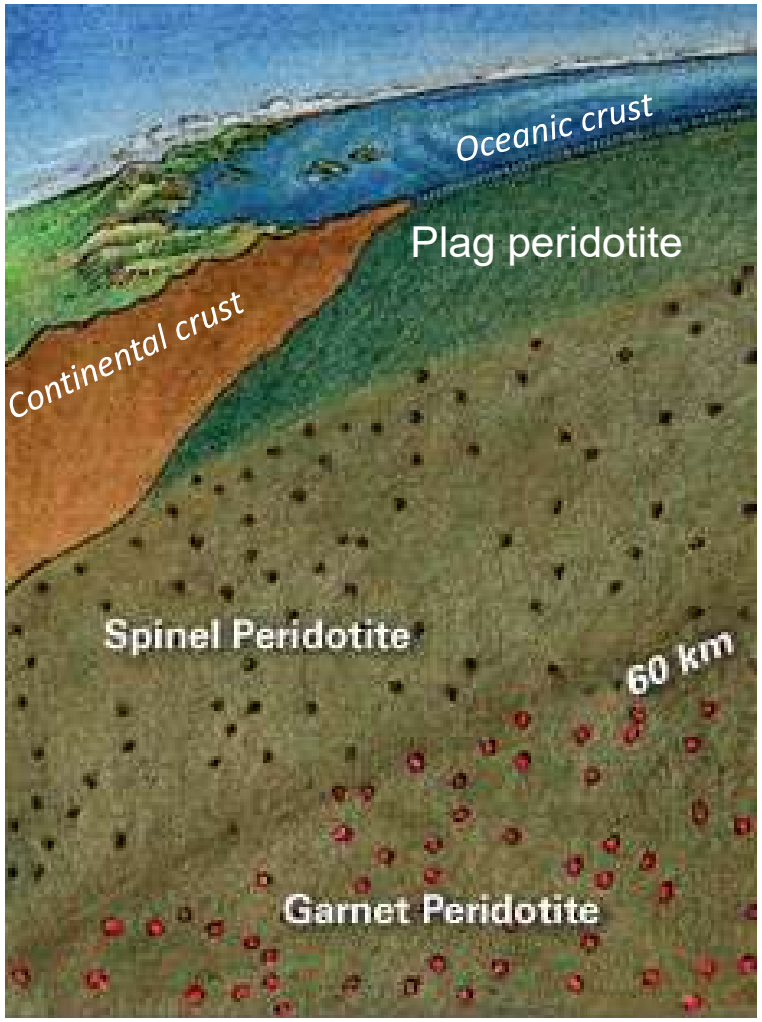
HOW ARE PARTITION COEFFICIENTS OF REE USEFUL?



When we consider the **origin of basalts**, we can use REE to:

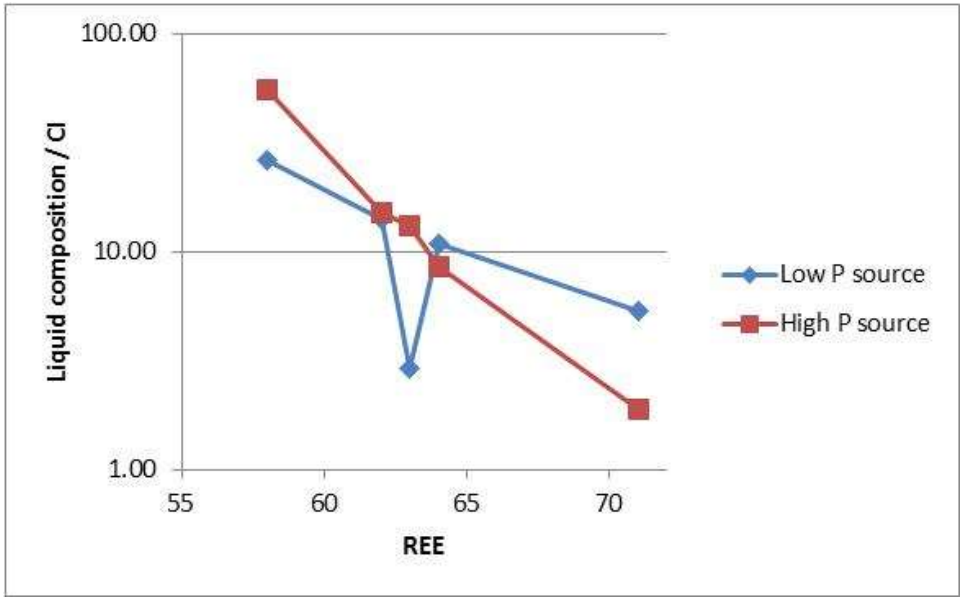
- Determine the extent of melting – did we move off the eutectic?
- Determine whether there has been any crustal assimilation as the basalt rose from the mantle – can we explain the REE abundance by mantle melting alone?
- Determine the depth of melting – what minerals were present in the mantle source?

DETERMINING THE DEPTH OF MELTING



Smithsonian Institution

Comparison of the REE abundance patterns expected for melts produced at low pressure and high pressure. (See practical.)



The REE pattern of a basalt depends on which minerals were present in the mantle source.

Chemistry of the Earth

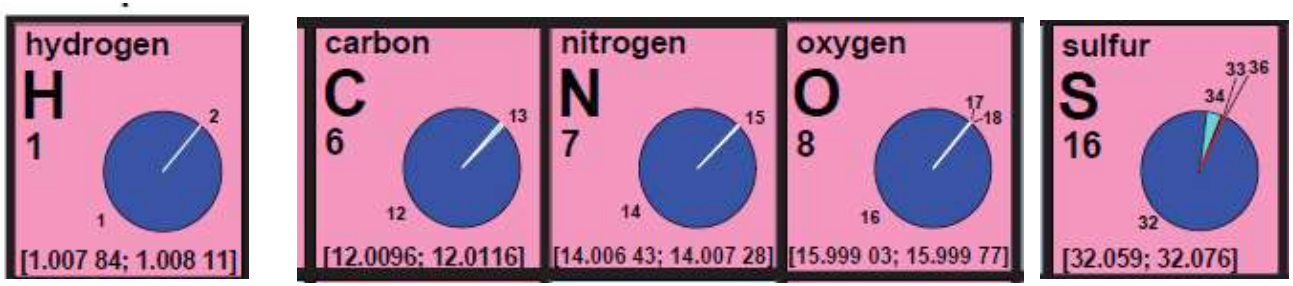
Lecture 8:
Stable Isotopes

OBJECTIVES OF LECTURE

- Investigate which elements provide the most useful stable isotopes for geochemistry, and why
- Understand how and why isotopes of the same element behave differently in geochemical systems
- Introduction to the terminology for stable isotope systems (α, δ)
- Look at how stable isotopes are used as geothermometers to interpret geochemical processes

OBJECTIVES OF PRACTICAL

- Understand the terminology of stable isotope studies
- Investigate how stable isotope geothermometers are used to interpret formation of rocks and mineral deposits



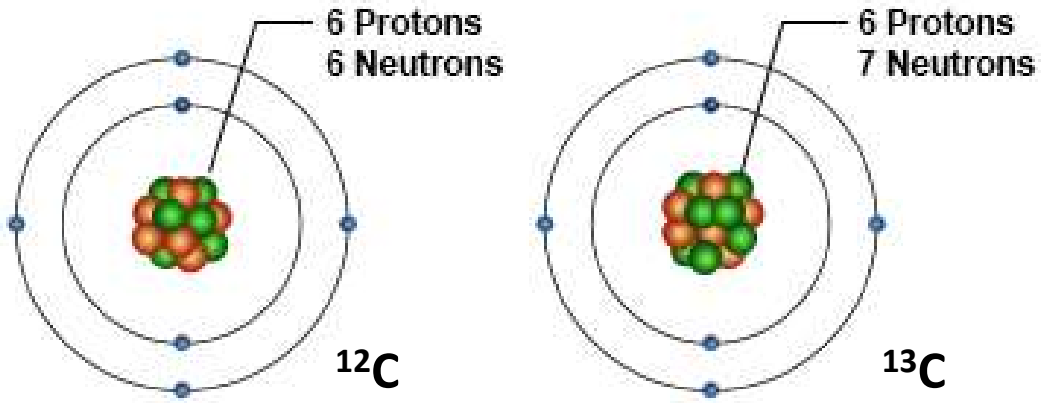
Why are these elements particularly useful for stable isotope studies?

1. The relative difference in mass between the different isotopes of these elements is large.
2. All these elements are abundant in nature and they are a major portion of common Earth materials. They all occur in common solids, liquids and gases.
3. C, N and S occur in different oxidation states.
4. All five elements have variable bonding in different materials.
5. The less abundant isotopes occur at measurable levels, making mass spectrometer analyses easier.
6. All five elements are biogenic, i.e. they are important elements in biologic materials.



Why do isotopes of the same elements behave differently?

Isotopes of the same element have the same electron configuration, so we might expect them to have the same chemical behaviour.



However, two different types of behaviour result in isotopic fractionation:

<u>Thermodynamic (equilibrium) mass fractionation</u>

Bonding energies between atoms are different depending on the mass of the atom that is bonding.

<u>Kinetic mass fractionation</u>

Kinetic energy of an atom or molecule is a function of mass (K.E. = ½ m v^2).

Important terminology for stable isotopes:

Fractionation factor (α)

Describes the distribution of stable isotopes between two phases.

For phases A and B, $\alpha_{A\text{-}B} = R_A / R_B$

- where R is the ratio of the heavy to light isotope.

This is a similar concept to a partition coefficient.

Example: to describe the fractionation of oxygen isotopes ^{16}O and ^{18}O between quartz and magnetite:

$$\alpha_{qz\text{-}mt} = (^{18}O/^{16}O)_{quartz} / (^{18}O/^{16}O)_{magnetite}$$



Quartz

=



Magnetite

When these minerals are in equilibrium, they have different $^{18}O/^{16}O$ ratios. This is because the bonds to O are different in the two minerals.

Important terminology for stable isotopes:

Delta notation (δ)

It is difficult to measure absolute values of isotope ratios, so measured values are compared to a standard.
The delta notation is a way of comparing an isotope ratio to the ratio measured in a standard.

$$\delta = (R_{sample} - R_{standard}) / R_{standard} * 1000 \quad ‰$$

Or…… $\delta = ((R_{sample} / R_{standard}) - 1) * 1000 \quad ‰$

Per mil ‰
Parts per thousand

Per cent %
Parts per hundred

Putting δ together with α:

$10^3 \ln\alpha$ is approximately the same as $\delta_A - \delta_B$

$$10^3 \ln\alpha \approx \delta_A - \delta_B = \Delta_{A\text{-}B}$$

Stable isotopes: Geothermometry using mineral pairs

At high temperatures, $1000 \ln \alpha_{A\text{-}B} \approx A / T^2$ (T in Kelvin)

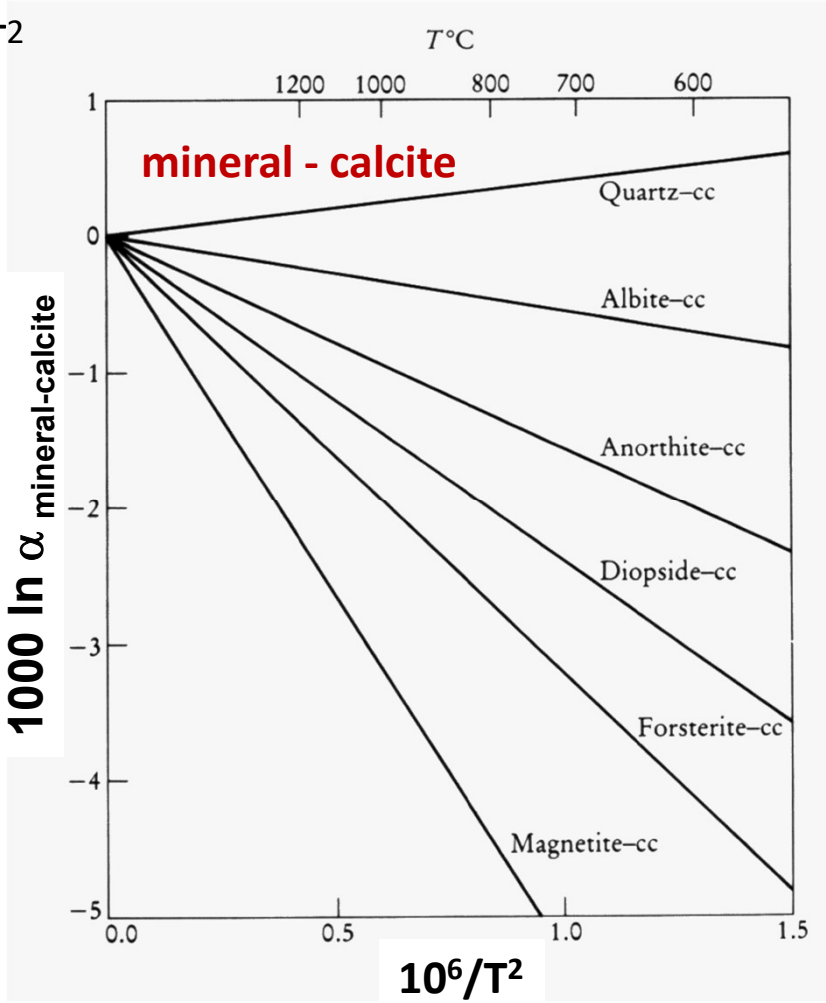
i.e. a straight line on a plot of $1000 \ln \alpha_{A\text{-}B}$ vs. T^2

The temperature dependence of α is the basis for stable isotope geothermometry.

At high temperatures, α approaches a value of 1, i.e. no fractionation (ln1 = 0).

So this technique works best for low-temperature equilibria where fractionations are larger.

Stable isotope geothermometry mostly uses isotopes of O and S.



Chemistry of the Earth

Lecture 9:
Water on the Earth

OBJECTIVES OF LECTURE

- Investigation of how stable isotopes are used to understand global meteoric water and palaeoclimate
- Examine how to describe the chemistry of water
- Investigate geochemical weathering reactions involving rocks and water

OBJECTIVES OF PRACTICAL

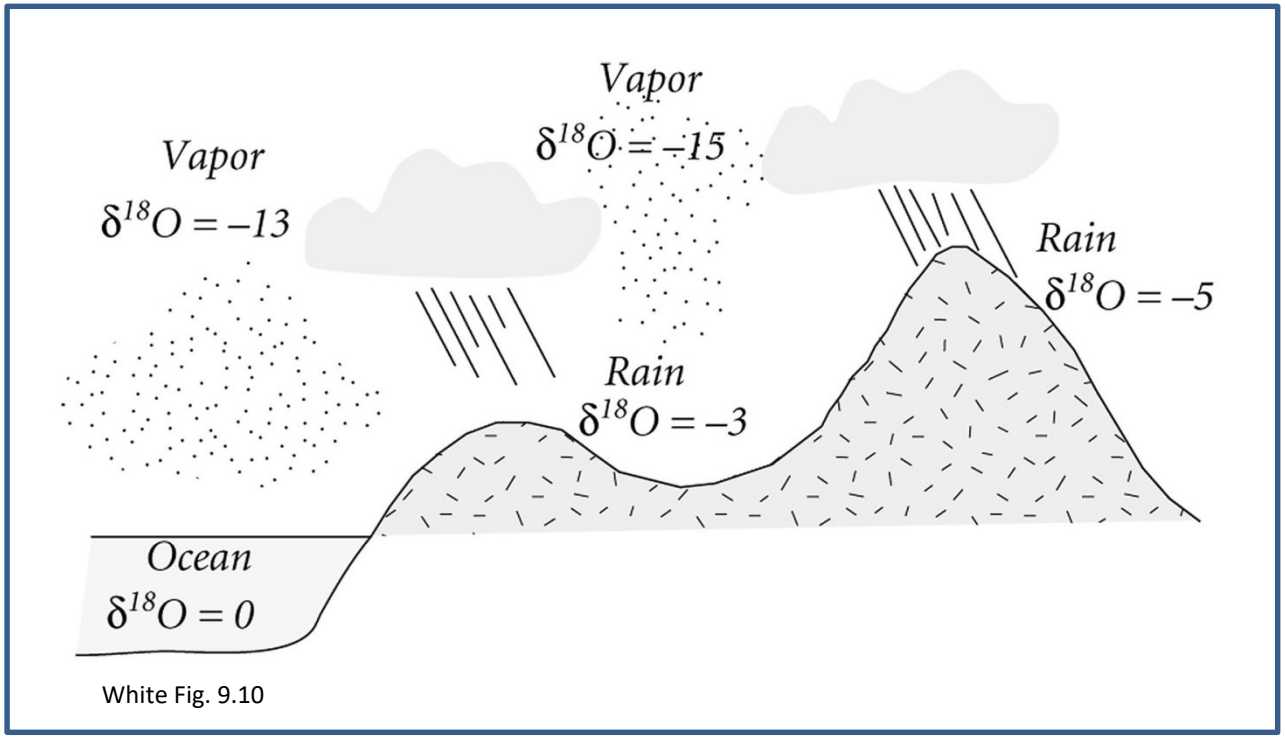
- Familiarization with the way chemical reactions are written for aqueous (electrolyte) solutions
- Understand why natural waters can be acidic
- Examine chemical weathering reactions to understand how rocks at the Earth's surface break down through chemical mechanisms

Evaporation and Condensation of Water: Oceans and Rain

Water, H_2O. Isotopes: ^{1}H, ^{2}H (=D), ^{16}O, ^{17}O, ^{18}O. Nine isotopologues.

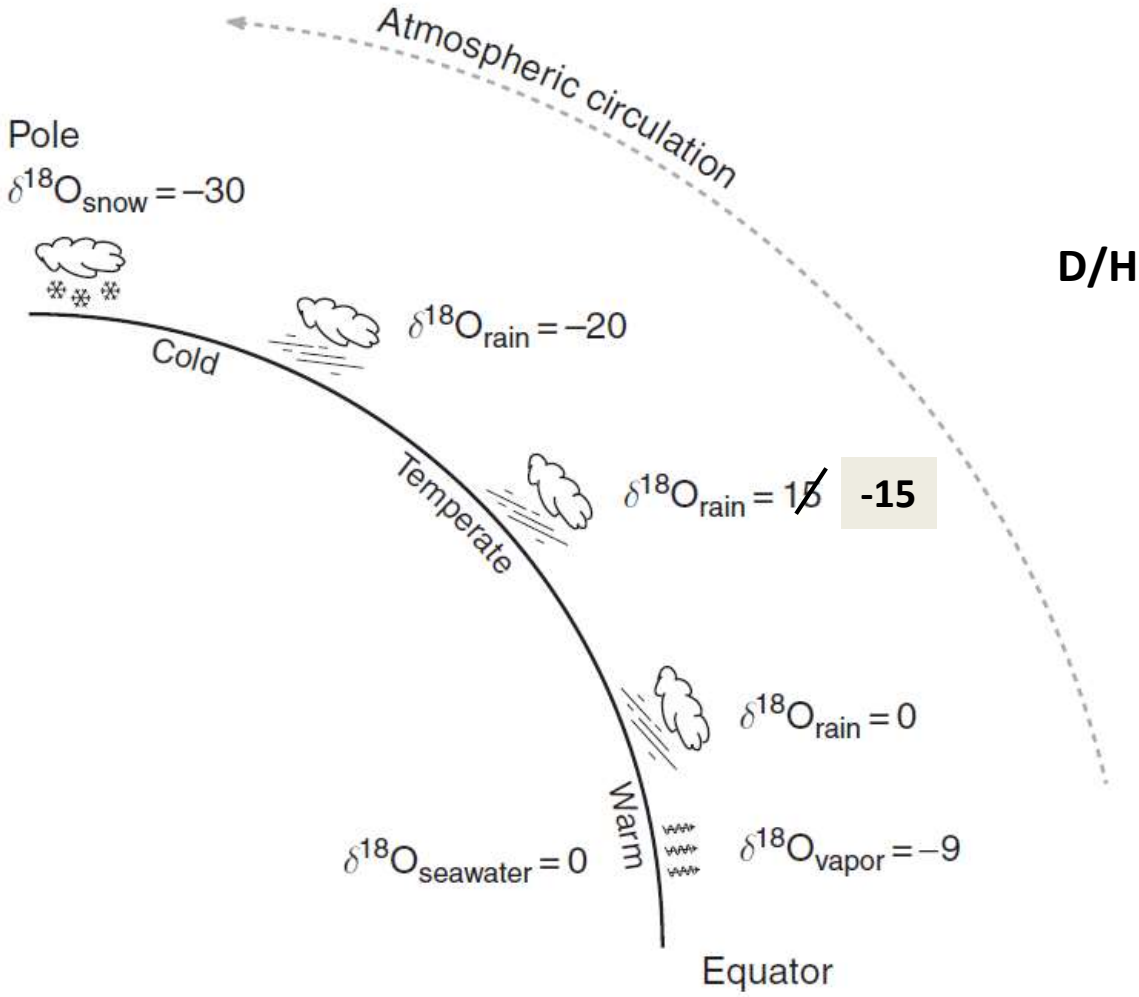
Ocean water: The ocean is well mixed and has a uniform isotopic composition. Mean values are very close to SMOW, i.e. δD and $\delta^{18}O$ are 0 ‰.

The ocean is an open system with respect to water vapour, which escapes to the atmosphere as soon as it forms. Isotopic fractionation occurs between water vapour and meteoric water (rain / snow).



White Fig. 9.10

Variation of $\delta^{18}O$ with latitude



D/H also varies with latitude

Albarede Fig. 7-9

Palaeoclimatology

During a glacial period, ice sheets lock up a disproportionate amount of ^{16}O, leaving the oceans relatively enriched in $\delta^{18}O$.

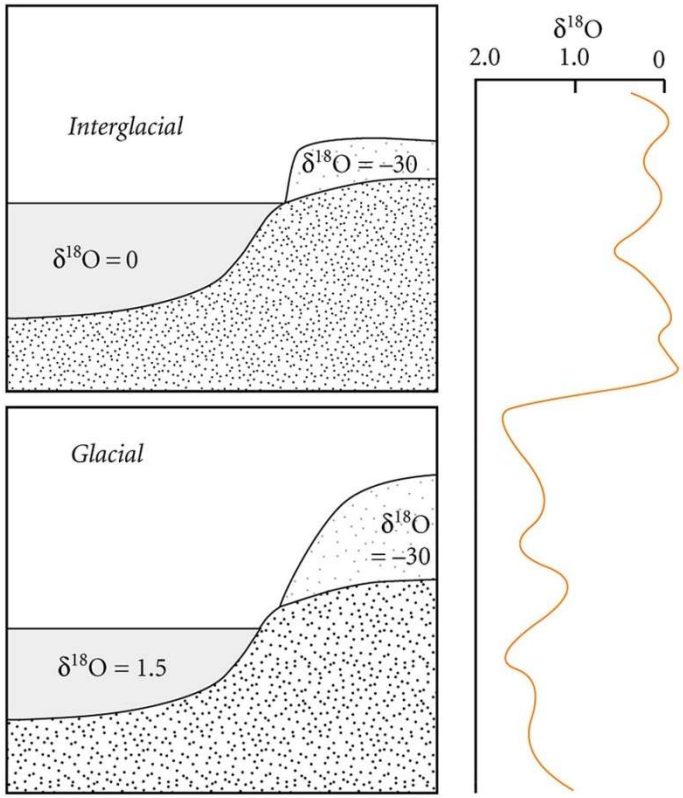
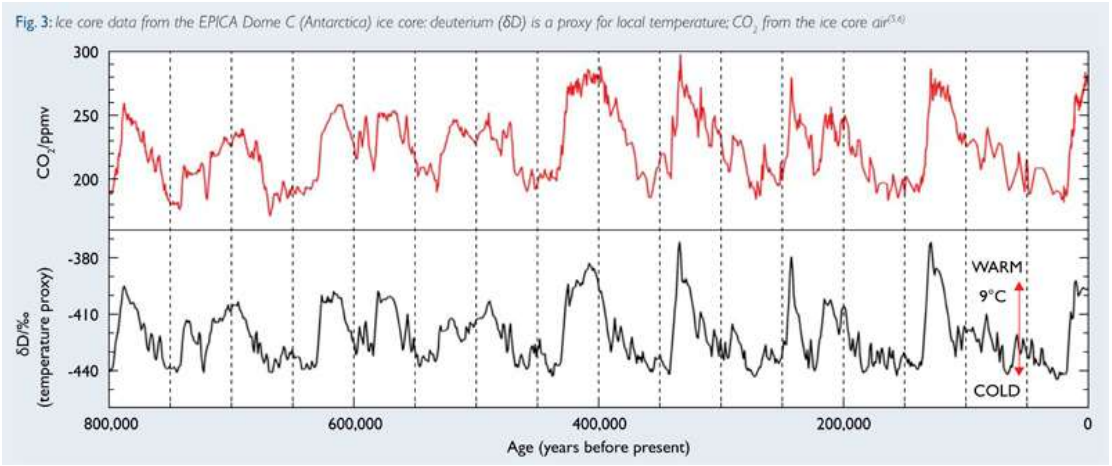


Figure 9.22 Cartoon illustrating how $\delta^{18}O$ of the ocean changes between glacial and interglacial periods.

Ice cores in Greenland and Antarctica preserve an important record of climate change: stable isotopes are used as a recorder of ocean temperature



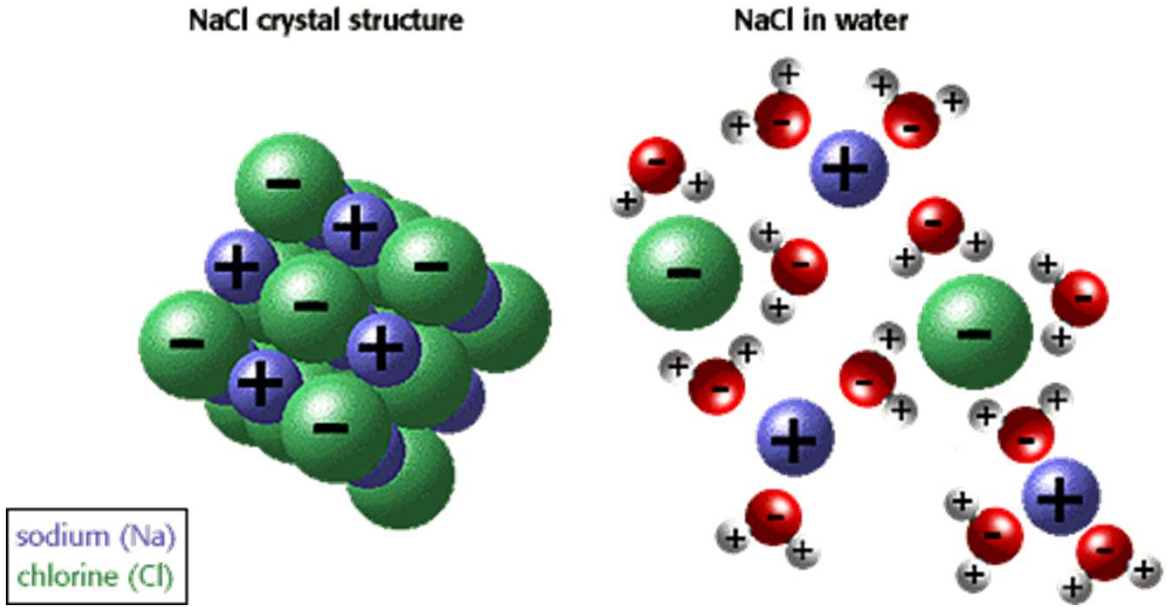
White, Fig. 9.22

WATER CHEMISTRY

When a **salt** dissolves in water, ions are surrounded by water molecules that orientate themselves with the opposite charge. This produces a layer of water that keeps the dissolved ions apart.

Example, halite: $NaCl_{(s)} = Na^{+}_{(aq)} + Cl^{-}_{(aq)}$

(aq) means the ion is in solution, (s) = solid



Water is the **SOLVENT**, NaCl is the **SOLUTE.**
A **SOLUTION** is the combination of both.

http://butane.chem.uiuc.edu/pshapley/genchem1/l21/1.html

Acidity and alkalinity

Misra Ch7.6
Misra p. 54

Acidity is a measure of the concentration of H^+ ions (protons) in solution.

pH means "potential" H.

pH = $-\log_{10}$ $[H^+]$, where $[H^+]$ is the concentration of H^+ ions in mol kg^{-1} (= mol l^{-1}).

$pH = -\log [H^+] = \log (1/[H^+])$

Lower pH indicates an acidic solution, higher pH an alkaline solution.

At 25 °C, pure water has a pH of 7.

Example: a solution contains 5×10^{-6} mol kg^{-1} hydrogen ions.

$$pH = \log (1 / (5 \times 10^{-6}))$$

$$pH = 5.30$$

{In fact, H^+ ions do not exist as isolated ions. They interact with water molecules to form hydronium ions, $H_3O^+_{(aq)}$}

WATER CHEMISTRY

Acidity

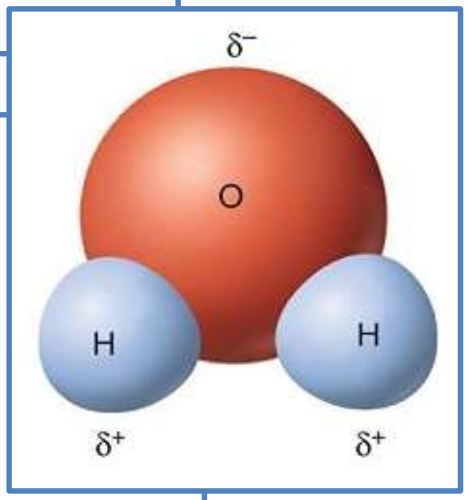
$pH = -\log [H^+] = \log (1/[H^+])$

Solubility product

For halite (sodium chloride),

$$NaCl_{(s)} = Na^+ + Cl^-$$

$$K_{sp} = [Na^+]\,[Cl^-]$$



Equilibrium Constant

For the reaction: $CO_{2\,(g)} + H_2O_{(l)} = H_2CO_{3\,(aq)}$

The equilibrium constant K is:

$$K = \frac{[H_2CO_3]_{(aq)}}{[CO_2]_{(g)}[H_2O]_{(l)}} \qquad K = \frac{A}{B * C}$$

Equations for equilibrium constants can be rearranged just like any algebraic equation:

$$[H_2CO_3]_{(aq)} = K_1 * [CO_2]_{(g)} * [H_2O]_{(l)} \qquad A = K * B * C$$

CARBONIC ACID AND CHEMICAL WEATHERING

Misra Ch7.5
Misra p. 53

Carbonic acid – an important naturally occurring acid that plays a critical role in chemical weathering of rocks.

CO_2 in the atmosphere dissolves in water to form carbonic acid:

$$CO_{2\ (g)} + H_2O_{\ (l)} = H_2CO_{3\ (aq)}$$

Carbonic acid is a weak acid. It dissociates in two steps:

$H_2CO_{3\ (aq)} = H^+ + HCO_3^-$ Bicarbonate: HCO_3^-

$HCO_3^- = H^+ + CO_3^{2-}$ Carbonate: CO_3^{2-}

- Rainwater in equilibrium with atmospheric CO_2 is acidic.
- Higher levels of CO_2 make rainwater more acidic.

Other acids in the atmosphere contribute to "acid rain", Including:

SO_2 gas →→ H_2SO_4 (sulphuric acid)
NO_x gas →→ HNO_3 (nitric acid)



Lovecz | Wikimedia.org

Misra Ch7.5 and p. 53

Chemistry of the Earth

Lecture 10:

Radioactive isotopes and Geochronology

OBJECTIVES OF LECTURE

• Consider what the “age” of a rock means
• Review how radioactive isotopes decay
• Understand how radioactive isotopes are used to determine the age of a rock – geochronometry
• Investigate which elements and isotopes make useful geochronometers, and why

OBJECTIVES OF PRACTICAL

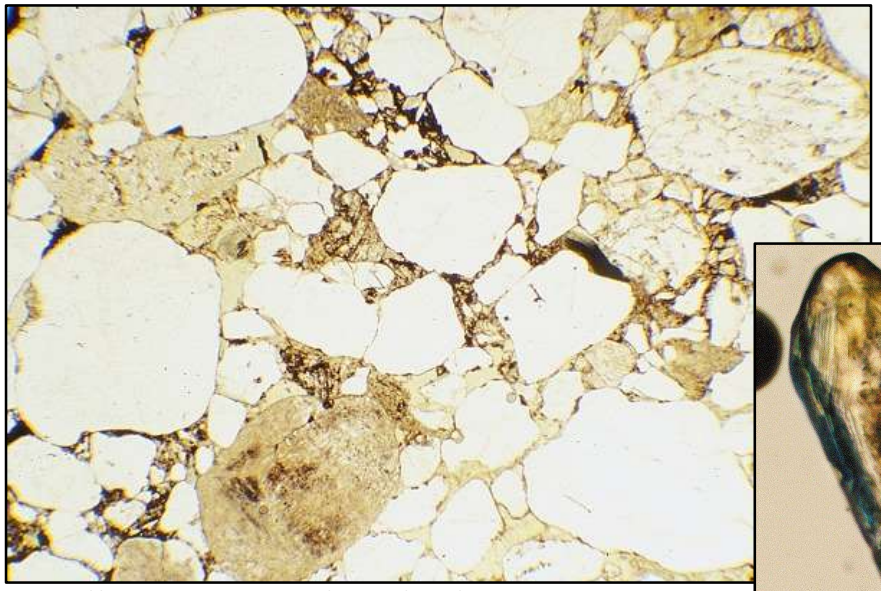
- Understand the principles of radioactive decay
- Investigate how isochrons are used to determine the ages of rocks

Sedimentary rocks

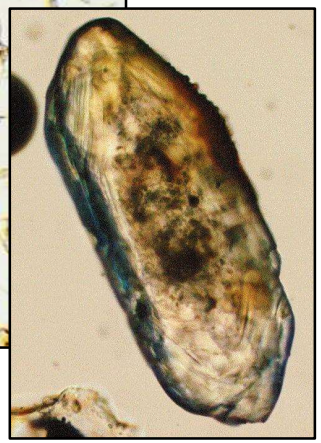
The meaning of an “age” for a sedimentary rock needs to be specified. Sedimentary rocks consist of grains from a range of different source rocks that each has its own age. The rock also has a cement that obviously was the last material to form.



Photograph copyright Bruce Watson, Rensselaer



https://www.earth.ox.ac.uk/~oesis/nws/nws-a98-25.html

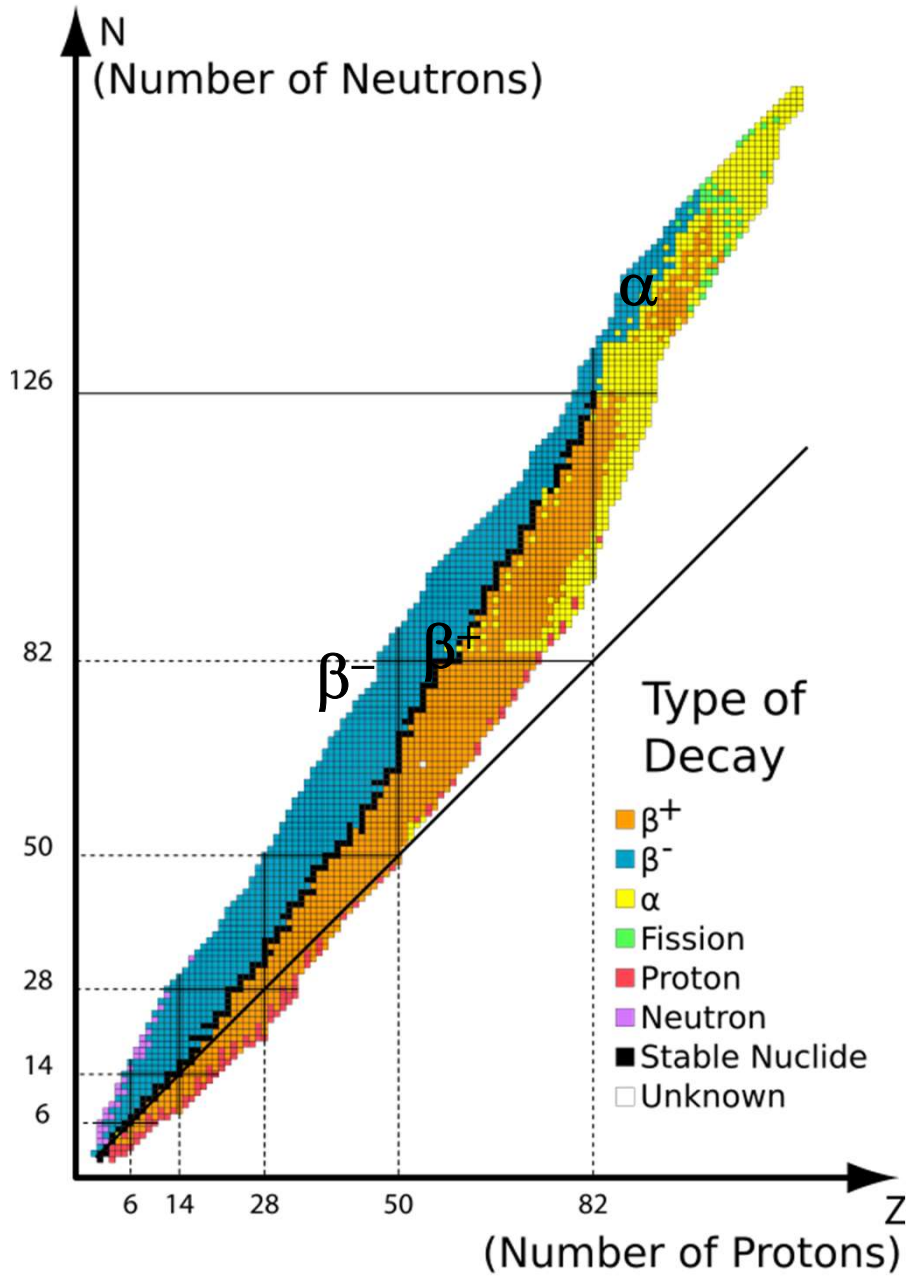


Zircon grain, 250 μm long.
Wikipedia

“The bits and pieces of rocks that make up the Jack Hills rock formation are ancient—over 3 billion years old. Individual crystals of zircon within the rocks are 4.4 billion years old, only 150 million years or so younger than the age of the Earth itself. These crystals are the oldest fragments of the Earth yet found.”

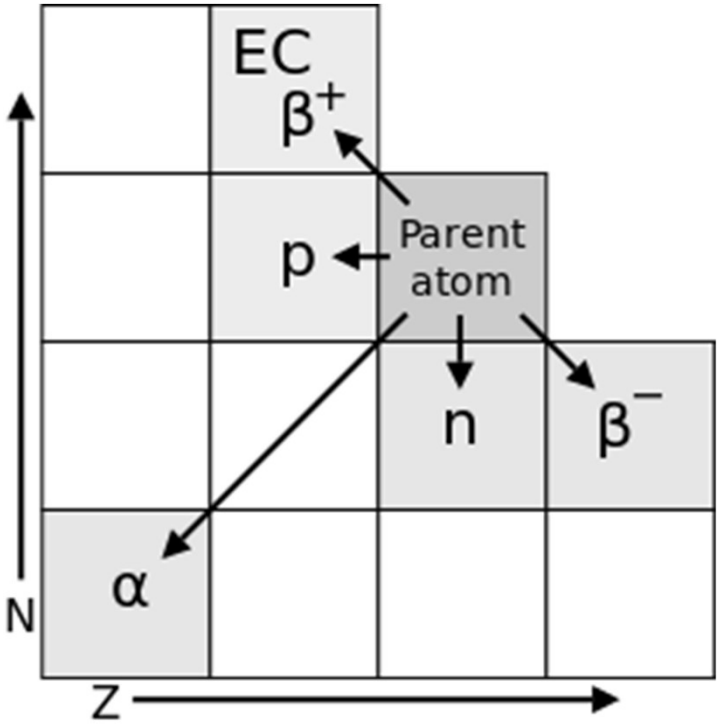
http://earthobservatory.nasa.gov/Features/Zircon/

Chart of the Nuclides



The most common types of decay are alpha (α) and beta (β) decay.

Different modes of radioactive decay lead different ways in the chart. Decay pathways lead to stable nuclides.



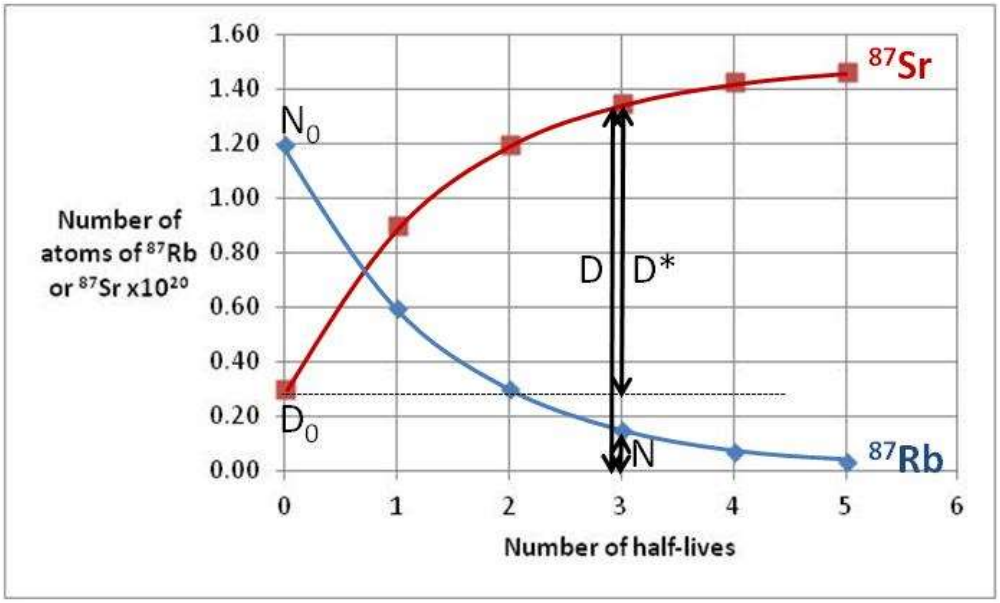
Basic equations for radioactive decay:

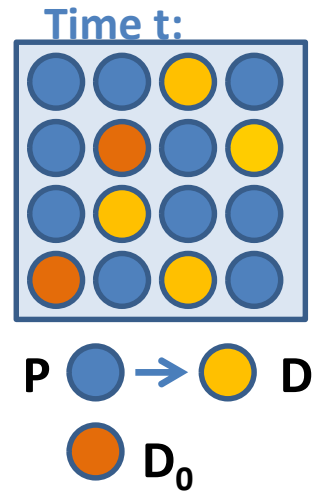
$$N / N_0 = e^{-\lambda t}$$

This equation relates the number of radioactive (parent) atoms (N), at any given time t, to the starting number of atoms (N_0 at t = 0) and the decay constant, λ.

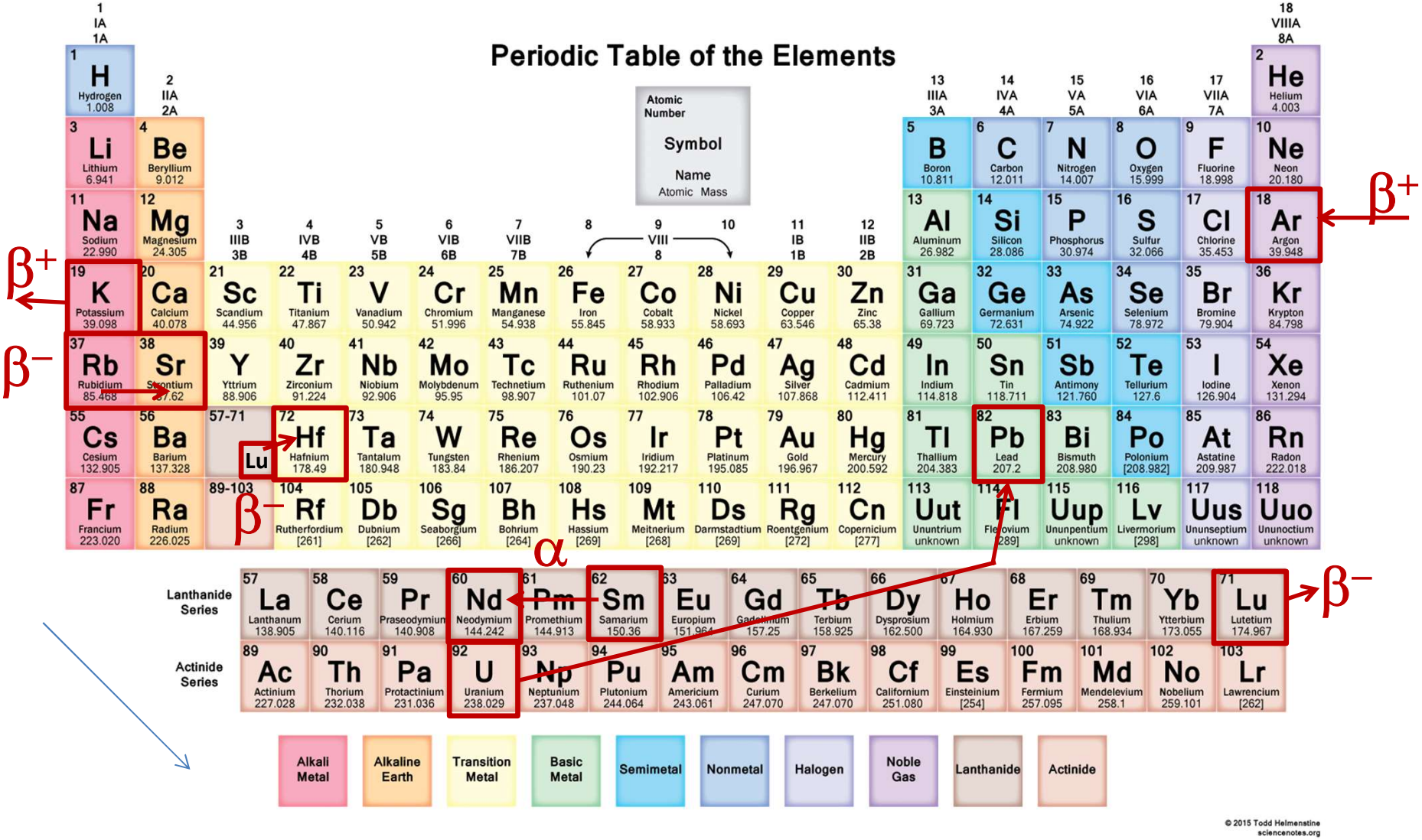
$$D = D_0 + N (e^{\lambda t} - 1)$$

Half-life and decay constant: $t_{½} = 0.693 / \lambda$

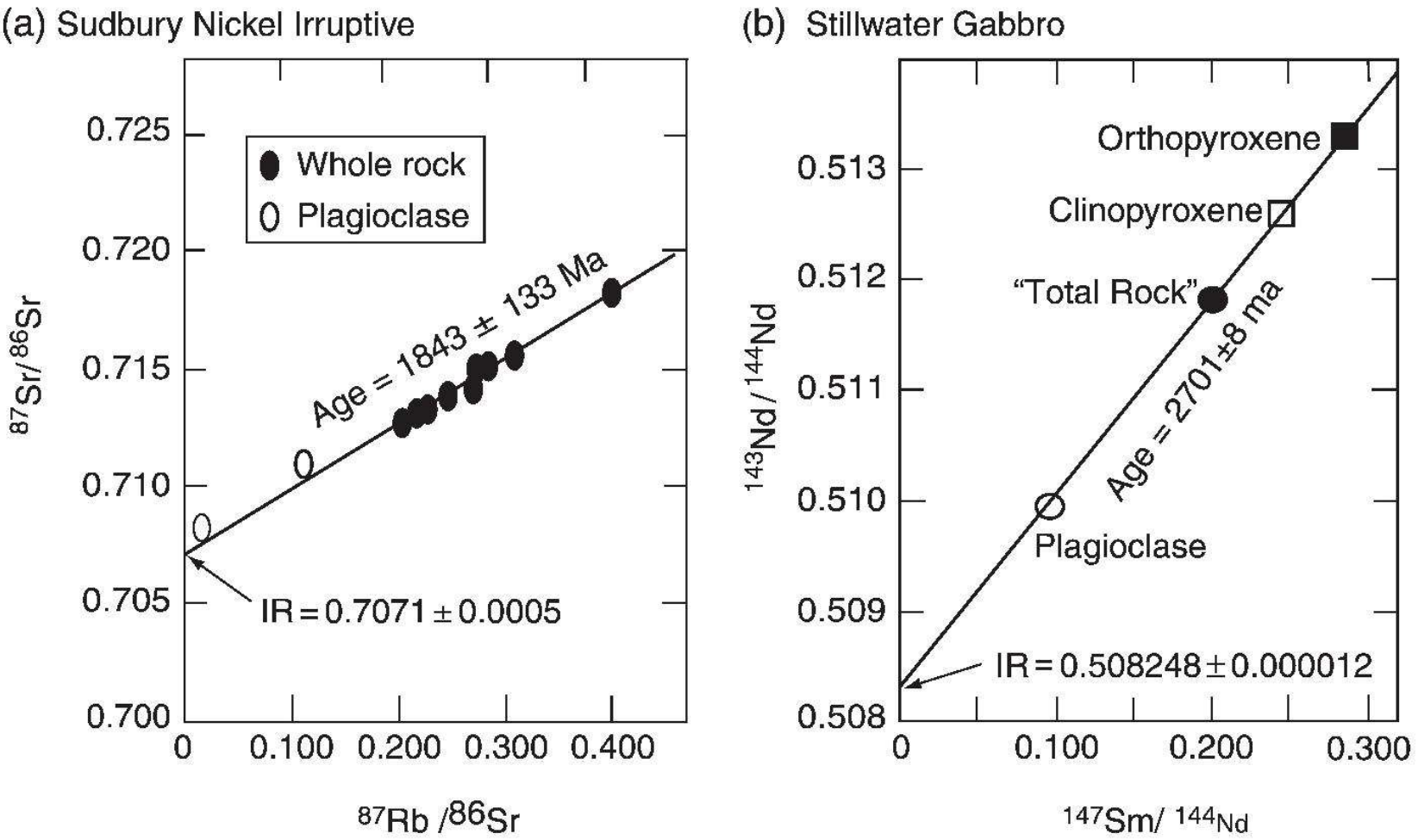




Important radioisotopes used for geochronometry



Isochrons



Rb-Sr isochron

Sm-Nd isochron

Misra, Figure 10.5

Rubidium-Strontium (Rb-Sr) system

Rubidium decay: ^{87}Rb to ^{87}Sr (β^-)

$$\left[\frac{^{87}Sr}{^{86}Sr}\right]_m = \left[\frac{^{87}Sr}{^{86}Sr}\right]_0 + \left[\frac{^{87}Rb}{^{86}Sr}\right]_m (e^{\lambda t} - 1)$$

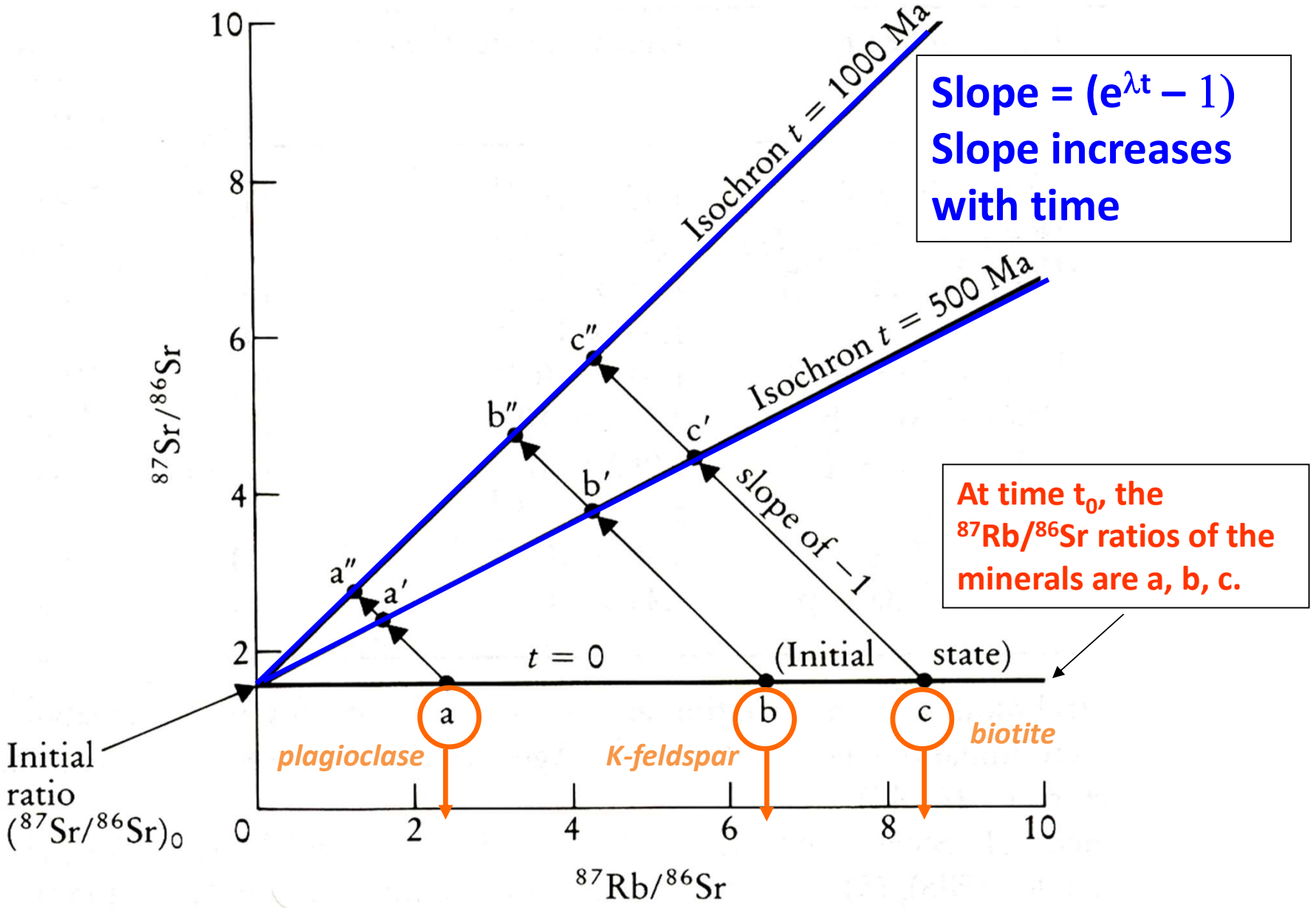
Measured with TIMS or ICP-MS: y axis

Intercept on y axis

Measured with TIMS or ICP-MS: x axis

Slope of the line on an isochron plot: allows us to determine t

$$y = c + x\,m$$


Slope = (e^λt – 1)
Slope increases with time
At time t0, the 87Rb/86Sr ratios of the minerals are a, b, c.
Isochron t = 1000 Ma
Isochron t = 500 Ma
slope of –1
t = 0
(Initial state)
a
b
c
plagioclase
K-feldspar
biotite
Initial ratio (87Sr/86Sr)0
87Sr/86Sr
87Rb/86Sr
0
2
4
6
8
10

Rollinson